AUTOMOBILE ENGINEERING · PART 12 OF 12

The Complete Series

By Isagi · 10 November 2026 · All specs and market figures as of November 2026

This is the whole Automobile Engineering series in one file: eleven parts, from how an engine works to how to buy a car, followed by a verification log. Page numbers below are for this file. Every part is also available on its own.

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Figures in Indian units (bhp, Nm, lakh, crore). Where a number could not be verified it carries an asterisk and the part's footnote says why.

AUTOMOBILE ENGINEERING · PART 1 OF 12

The Engine, Part 1: How It Works

By Isagi · 9 November 2026 · All specs as of November 2026 (manufacturer brochures and listings)

TL;DR

An engine is an air pump that burns fuel to push pistons. Each push turns a crankshaft, and that rotation eventually reaches the wheels. **CC** (cubic centimetres) is simply the total volume the pistons sweep: a measure of how much air the engine can breathe per cycle, not of how much power it makes. Power depends on how much air gets in, how much fuel is burnt with it, and how fast the engine spins. That is why a 1.2-litre turbo can out-pull a 1.5-litre non-turbo.

1. Where the engine sits in the car

In a typical Indian petrol or diesel car the engine is a single block of metal under the bonnet, bolted to a gearbox. Everything else in the car (suspension, brakes, electronics) exists to carry, steer and stop what the engine moves. For a manager it is the single costliest mechanical part and the one most shaped by regulation (BS6 emission norms) and tax (engine size and fuel type affect the slab a car falls in).

2. The parts that matter

Five parts explain nearly everything:

- **Cylinder:** a sealed chamber where fuel burns. Cars have 3 or 4 in India; a few have more.
- **Piston:** a plug inside the cylinder, pushed down by the burning gas.
- **Connecting rod and crankshaft:** convert the piston's up-down motion into rotation.
- **Valves:** doors at the top of the cylinder. Intake valves let air (and fuel) in; exhaust valves let burnt gas out.
- **Spark plug (petrol) or fuel injector (diesel):** starts the combustion.

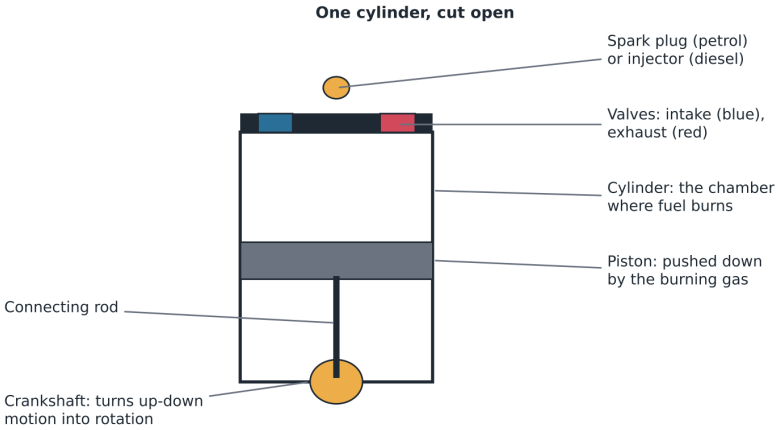


Fig 1. One cylinder, cut open. Everything else in the engine is repetition and refinement of this.

3. The four-stroke cycle

Almost every car engine repeats the same four-step loop thousands of times a minute. A *stroke* is one trip of the piston up or down, so one full cycle takes four strokes and two turns of the crankshaft.

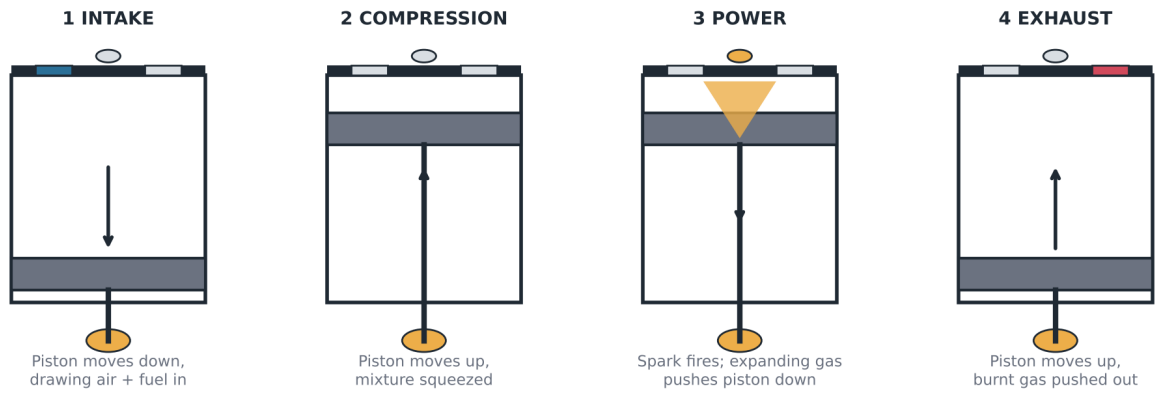


Fig 2. Intake, compression, power, exhaust. Only the third stroke produces work; the other three are paid for by the crankshaft's momentum (and by other cylinders).

- **Intake:** piston moves down; the intake valve opens and air (plus fuel, in a petrol engine) is drawn in.
- **Compression:** both valves shut; the piston moves up and squeezes the mixture. More squeezing means more energy per burn, which is why compression ratio matters.
- **Power:** a spark (petrol) or the heat of compressed air igniting injected fuel (diesel) triggers combustion; the expanding gas drives the piston down.
- **Exhaust:** the exhaust valve opens and the rising piston pushes burnt gas out.

4. What CC actually means

Each cylinder's **swept volume** is the area of the piston face times the distance it travels (the stroke): **π/4 × bore² × stroke**. Multiply by the number of cylinders and you get engine capacity.

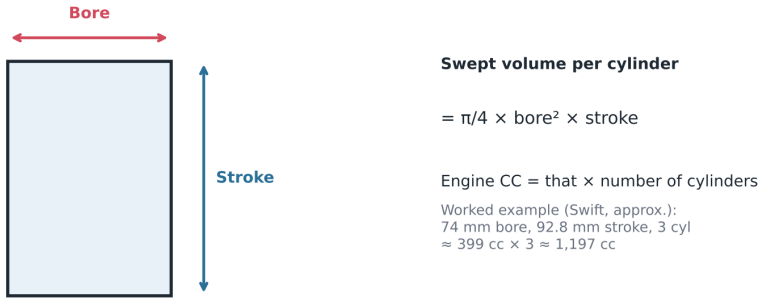


Fig 3. Bore is the cylinder's width; stroke is the piston's travel. Swift figures are third-party database values and should be treated as approximate.

Take the Maruti Suzuki Swift's Z12E engine: 1,197 cc across 3 cylinders, so each cylinder sweeps about 399 cc. It makes 80.46 bhp at 5,700 rpm and 111.7 Nm at 4,300 rpm.

Same CC, very different engines. The Tata Nexon's 1.2 Revotron turbo is 1,199 cc, effectively identical to the Swift's. Yet it makes 118.27 bhp and 170 Nm, about 47% more power and 52% more torque from the same cylinder volume. The reason is the turbocharger: it packs more air into each cylinder, so each cycle burns more fuel. CC measures the size of the lungs, not the athlete.

CC and tax. India has historically taxed cars partly by engine size and length (the sub-1,200 cc petrol / sub-1,500 cc diesel, sub-4 m category was the benefit slab). Rates changed with the September 2025 GST revision, so check the current slab before using engine size as a tax shortcut.*

5. Cylinders: why 3, 4 or 1

For a given CC, more cylinders means smaller, lighter pistons that fire more often, which gives a smoother, quieter engine that revs freely. Fewer cylinders means fewer parts, less friction and lower cost, but more vibration. Indian small cars mostly use three cylinders for cost and efficiency; four-cylinder engines appear where refinement is a selling point (the Creta's 1.5 MPi is a 4-cylinder, 1,497 cc, 113 bhp unit).

The Swift's move from the older K12 four-cylinder to the three-cylinder Z12E shows the trade-off: lighter, cheaper and more efficient, in return for a slightly less smooth idle. **Valves** add a further lever: most modern engines use four valves per cylinder (two intake, two exhaust) because more door area lets more air in and out.

6. Petrol versus diesel

Both use the four strokes; they differ in how combustion starts. A petrol engine mixes fuel with air and ignites it with a spark. A diesel engine compresses air alone, far harder (compression ratio roughly 16–18:1 versus roughly 10–13:1 for petrol), until it is hot enough to ignite fuel sprayed in.

Diesel's high compression gives more torque at low rpm and better fuel economy, but the engine is heavier, and BS6 emission control (particulate filter, urea-based SCR) adds cost. The Creta shows it clearly: the 1.5 diesel (1,493 cc) makes 114 bhp but 250 Nm from 1,500 rpm; the 1.5 petrol MPi (1,497 cc) has almost the same cc and 113 bhp, but only 143.8 Nm. Same size, similar power, 74% more torque.

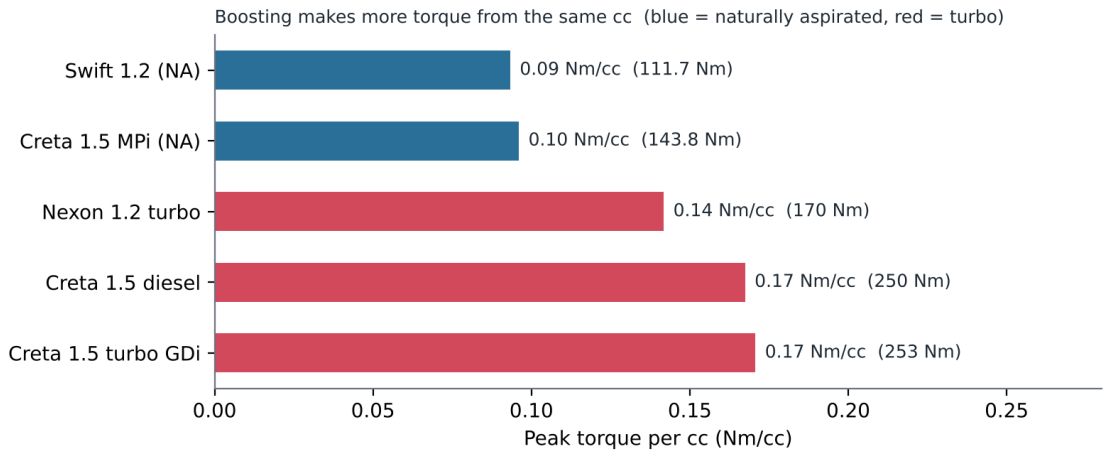


Fig 4. Torque per cc for five engines in this article. Boosted engines (red) deliver roughly 0.14–0.17 Nm per cc against about 0.09–0.10 for the plain petrols (blue).

7. The Indian engine line-up in one table

Car / engine	CC	Cyl	Power (bhp)	Torque (Nm)	Aspiration
Swift 1.2 Z12E	1,197	3	80.46 @ 5,700	111.7 @ 4,300	Natural
Nexon 1.2 Revotron	1,199	3	118.27 @ 5,500	170 @ 1,750–4,000	Turbo
Creta 1.5 MPi	1,497	4	113 @ 6,300	143.8 @ 4,500	Natural
Creta 1.5 diesel	1,493	4	114 @ 4,000	250 @ 1,500–2,750	Turbo diesel
Creta 1.5 turbo GDi	1,482	4	158 @ 5,500	253 @ 1,500–3,500	Turbo
Thar 2.0 mStallion	1,997	4	~150	300–320	Turbo
Thar 2.2 mHawk	2,184	4	~130	~300	Turbo diesel

Five of these seven engines are turbocharged (Nexon, Creta diesel, Creta turbo, and both Tharas). They make about 0.14–0.17 Nm per cc against roughly 0.09–0.10 for the two naturally aspirated

petrols (Swift, Creta MPi). The modern trend is simple: shrink the engine, add a turbo, and keep the power.

8. Japanese and German engines in India

Japanese and German makers solve the same physics with different philosophies, and both are well represented here. Japanese brands (Maruti Suzuki, Toyota, Honda) dominate the mass market. German-group brands sit in the mid-market (Skoda and Volkswagen) and at the top (BMW, Mercedes-Benz, Audi).

Car / engine	Origin	CC	Cyl	Power	Torque (Nm)	Approach
Honda City 1.5 i-VTEC	Japan	1,498	4	~119 bhp (121 PS)*	145*	Natural, high-revving
Toyota Innova Hycross 2.0 petrol	Japan	1,987	4	~173 bhp	204	Natural
Toyota Innova Hycross 2.0 hybrid	Japan	1,987	4	~184 bhp system	~188 (engine)	Natural + electric motor
Skoda Kushaq / Slavia 1.0 TSI	Germany (VW Group)	999	3	115 PS	178*	Small turbo
Skoda Kushaq / Slavia 1.5 TSI	Germany (VW Group)	1,498	4	150 PS	250*	Turbo, cylinder deactivation*
BMW 330Li	Germany	1,998	4	258 hp	400	Turbo
Mercedes-Benz C 220d	Germany	1,993	4*	197 bhp	440	Turbo diesel + 48V mild hybrid

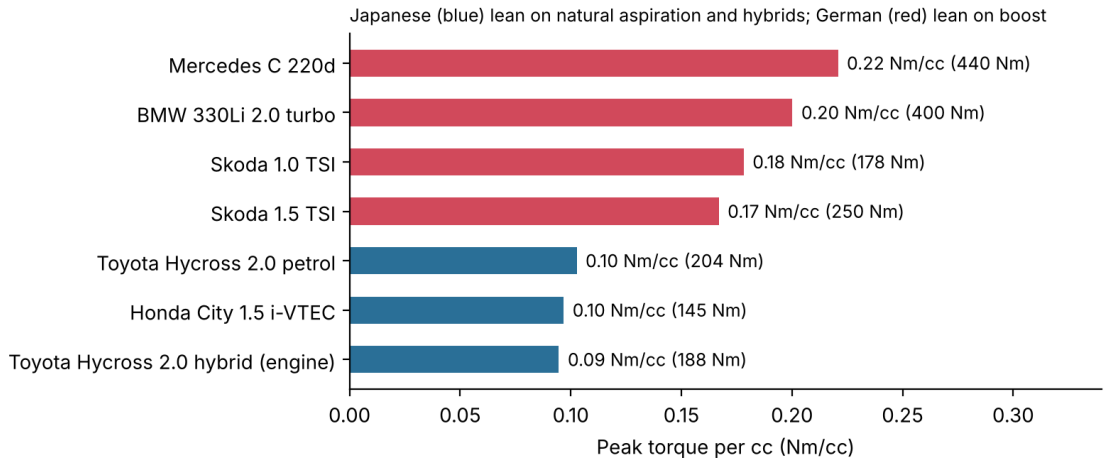


Fig 5. Torque per cc. Japanese naturally aspirated engines sit near 0.10 Nm per cc; German boosted engines run 0.17–0.22.

The Japanese way: breathe naturally, then add electric help. Honda's 1.5 i-VTEC makes about 0.10 Nm per cc with no turbo. It needs revs to deliver its power, but it is simple, smooth and famously durable. Toyota's 2.0-litre goes further: the hybrid version adds an electric motor to the same engine, chasing efficiency rather than peak torque. Maruti's Swift (Section 4) follows the same logic: small, light, cheap to run.

The German way: shrink the engine, add a turbo. Skoda's 1.0 TSI is a three-cylinder of just 999 cc, yet its torque per cc (0.18) is nearly double the Honda's. The 1.5 TSI can shut down two of its four cylinders at light load to save fuel.* BMW's 2.0-litre turbo petrol and Mercedes' 2.0 diesel reach 0.20 and 0.22 Nm per cc. Mercedes also adds a 48-volt mild-hybrid system that gives a short boost of up to 20 hp and 200 Nm.

Neither is simply better. Boosted engines pull hard at low rpm from small capacity, which suits city and highway driving, but run hotter and have more parts to maintain. Naturally aspirated engines are simpler and reward revs, but need more cc for the same pull. Part 2 explains how to choose between them for the way you drive.

Summary

- An engine burns fuel in cylinders; pistons turn a crankshaft; the crankshaft drives the wheels.
- Four strokes: intake, compression, power, exhaust. Only one produces work.
- CC = π/4 × bore² × stroke × cylinders. It measures size, not power.
- Same CC can mean very different outputs: Swift vs Nexon (1.2 L, 80 vs 118 bhp).
- 3 cylinders: cheaper and efficient; 4 cylinders: smoother.
- Diesel compresses air harder, giving more low-rpm torque but costlier BS6 hardware.
- Five of seven engines in our sample are boosted, which is the market's direction.
- Japanese makers lean on natural aspiration and hybrids (about 0.10 Nm/cc); German makers lean on small turbos (0.17–0.22 Nm/cc).

*Unverified or approximate: GST thresholds (18% versus 40% from 22 September 2025); Swift Z12E bore, stroke and compression ratio from third-party databases; compression-ratio ranges and efficiency rules of thumb are textbook generalisations. Skoda 1.0/1.5 TSI torque and cylinder deactivation, Honda City figures and the Mercedes C 220d cylinder count are from listings or general knowledge, not an official brochure. Specs: manufacturer brochures, press kits and listings, November 2026. Theory: Heywood, Internal Combustion Engine Fundamentals.

See also: Automobile Engineering 2/12 — The Engine, Part 2: power, torque and mileage.

AUTOMOBILE ENGINEERING · PART 2 OF 12

The Engine, Part 2: Power, Torque and Mileage

By Isagi · 9 November 2026 · All specs as of November 2026 (manufacturer brochures and listings)

TL;DR

Power is how fast an engine can do work; torque is how hard it can push. You feel torque when pulling away or climbing, and power when overtaking at speed. To pick a car, look at the **spread** of torque (the rpm range where it is available) and the car's power-to-weight, not the single biggest number. Mileage is decided by how much energy it takes to move the car (weight, drag, tyres) and how efficiently the engine and gearbox turn fuel into that energy. ARAI figures are lab numbers; owners of the Creta variants report city figures roughly 17% to 46% lower.

1. Torque and power in plain words

Torque is the twisting force the engine puts on the crankshaft, measured in newton-metres (Nm). **Power** is the rate of doing work, which is torque multiplied by how fast the engine is turning. The conversion is worth knowing once:

Power (kW) = Torque (Nm) × rpm ÷ 9,549, and **bhp = kW ÷ 0.746**.

Use it on the Swift's spec sheet. Peak torque is 111.7 Nm at 4,300 rpm, which works out to about 50 kW or 67 bhp at that moment. The 80.46 bhp peak arrives later, at 5,700 rpm, where torque has slipped to about 100 Nm (60 kW × 9,549 ÷ 5,700). Torque falls at high rpm because the engine has less time to fill its cylinders, but rpm keeps rising, so power keeps climbing for a while longer.

Illustrative curves: the shape is typical; only the marked points come from the spec sheets

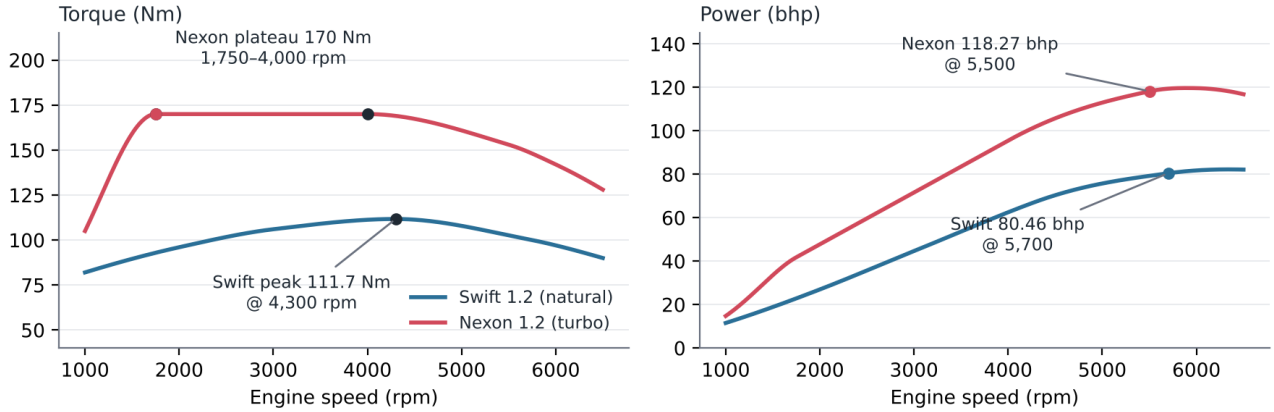


Fig 1. Torque (left) and power (right) against engine speed. The Nexon's turbo gives it a flat torque plateau from low rpm; the naturally aspirated Swift peaks later and lower.

2. Reading the numbers on a spec sheet

A spec line such as **118.27 bhp @ 5,500 rpm, 170 Nm @ 1,750–4,000 rpm** tells you three things: how much, where it peaks, and how wide the torque plateau is. The plateau matters most in daily driving.

- **Nexon 1.2 turbo:** 170 Nm is available from just 1,750 rpm to 4,000 rpm. The car pulls without you changing gear.

- **Creta 1.5 diesel:** 250 Nm from 1,500 to 2,750 rpm. At 2,750 rpm that is about 72 kW or 96 bhp, strong pull from barely above idle, which is why diesels feel effortless in traffic and on climbs.
- **Creta 1.5 MPi petrol:** 143.8 Nm only at 4,500 rpm. You need to keep revs up, so expect more downshifting.

Torque at the wheels is engine torque multiplied by the gear ratio, so gearing changes how a car feels as much as the engine does. Part 3 covers that.

3. How to pick a car by power

- **Start with your driving.** City crawl, mixed use, regular highway, hills, a full family load: each rewards a different number.
- **Look at the torque spread,** not just the peak. A wide plateau starting below 2,000 rpm means relaxed driving.
- **Check power-to-weight:** bhp divided by kerb weight in tonnes. The Swift's 80.46 bhp over a listed 925 kg is about 87 bhp per tonne.* As a rough guide, under 70 feels leisurely, 80 to 100 is brisk enough for most buyers, and above 120 is quick.*
- **Remember the gearbox.** An automatic or CVT can dull a small engine's response; a turbo petrol with a DCT feels sharper.
- **Don't pay for power you can't use.** Speed limits and traffic cap how much extra power you will ever use.

If you mostly...	Look for	Examples from our list
Drive in the city, 1–2 people	Light weight, easy low-rpm response, an easy automatic	Swift 1.2; Nexon 1.2 turbo (plateau from 1,750 rpm)
Drive mixed with a family of four	Torque available from about 1,500–2,000 rpm	Nexon 1.2 turbo; Creta 1.5 diesel
Do highway runs and overtake often	Power at speed, good power-to-weight, a tall top gear	Creta 1.5 turbo (158 bhp); Skoda 1.5 TSI (150 PS)
Climb hills or carry full loads	A wide torque plateau; diesel or turbo	Creta 1.5 diesel; Thar diesel (about 300 Nm)

4. What actually makes mileage better

Mileage depends on two things: how much energy the car needs, and how well the engine and drivetrain supply it.

Energy the car needs. Three forces resist motion. *Weight* matters whenever you accelerate or climb: more mass, more fuel. *Rolling resistance* comes from the tyres and rises with weight and low tyre pressure. *Aerodynamic drag* grows with the square of speed, so going from 100 to 120 km/h raises drag force by 44% and the power needed to beat it by about 73%. Speed is the biggest lever a driver controls.

How well the engine converts fuel. Textbooks put the best-case thermal efficiency of a petrol engine at roughly 30–35% and a diesel at roughly 40%; the rest leaves as heat.* Real driving is worse, because city driving uses only a small fraction of the engine's capability, where friction and pumping losses take a bigger share. That is why start-stop, cylinder deactivation and hybrids help most in the city.

Drivetrain and conditions. More gear ratios keep the engine in its efficient range. Air conditioning, traffic, tyre pressure, a loaded boot and a heavy foot all change the result.

5. ARAI versus real-world mileage

The ARAI figure comes from a standard test on a laboratory roller, with fixed speeds, no air conditioning and no traffic. It is good for comparing cars with each other, not for predicting your fuel bill. The Hyundai Creta shows the gap clearly:

Creta 1.5 variant	ARAI claim (km/l)	City, owner-reported	Highway, owner-reported	City shortfall
Petrol, manual	17.4	12–14	up to about 17	20–31%
Turbo petrol, DCT	18.4	10–12	16–17	35–46%
Diesel, manual	21.8	16–18	over 20	17–27%

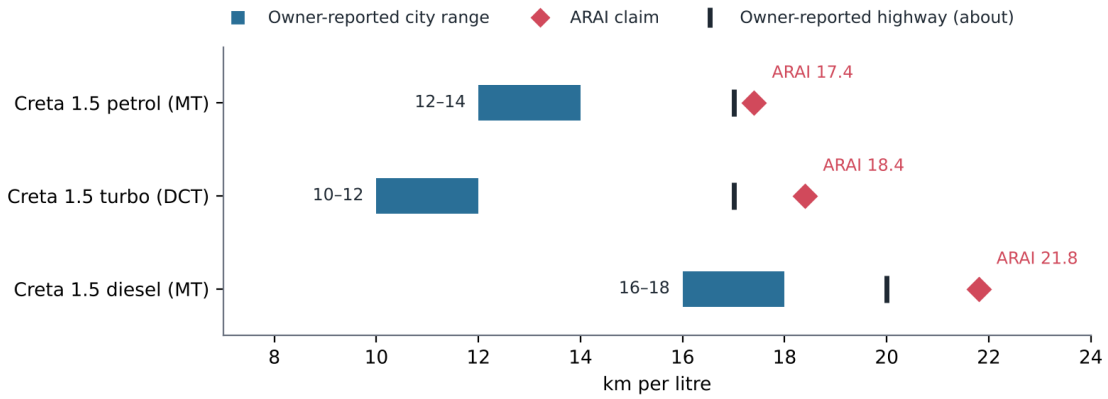


Fig 2. ARAI claim versus owner-reported figures for three Creta 1.5 variants. Owner-reported figures vary with traffic, AC use and driving style.

Note the turbo petrol: it has the best ARAI figure of the three petrols and the worst city figure. Boost helps in a gentle lab test and costs fuel when real traffic makes you use it.

6. Turbo, downsizing and hybrids

Downsizing replaces a big engine with a small turbocharged one (Section 8 of Part 1: Skoda's 1.0 TSI, Tata's 1.2 Revotron). A small engine at steady cruise runs at a higher load, which is more efficient, and the turbo supplies extra pull on demand. The weakness is the one in Fig 2: when you demand the boost, the engine burns more fuel.

Cylinder deactivation shuts some cylinders at light load; Skoda's 1.5 TSI is reported to do this on two of its four.* **Diesel** is more efficient than petrol at the same size, but costs more to buy and to keep BS6-compliant.

Hybrids change the picture most. The Honda City e:HEV claims about 27.3 km/l and the Toyota Hyryder strong hybrid about 28.0 km/l (ARAI), roughly 57–61% more than the petrol Creta's 17.4.* These are different cars, so the comparison shows scale rather than an exact gain. Part 7 covers hybrids properly.

7. BS6 and E20: what changed for engines

India moved to **BS6** emission norms on 1 April 2020, and to a stricter second phase on 1 April 2023 that added real-driving emission testing and on-board diagnostics.* For petrol engines this mostly meant better catalytic converters and engine tuning. For diesels it meant a particulate filter and a urea-based SCR system, which add cost, and some makers, notably Maruti Suzuki, stopped selling small diesels altogether.*

E20 petrol (20% ethanol) is now the standard pump fuel nationwide. Ethanol carries less energy per litre than petrol. The petroleum ministry's estimate (August 2025) is a mileage loss of about

1–2% for E10-designed cars calibrated for E20 and 3–6% for others; the transport minister told Parliament in July 2026 that the drop is 2–6%.

Summary

- Torque is push; power is push times speed. kW = Nm × rpm ÷ 9,549, and bhp = kW ÷ 0.746.
- Peak torque comes at lower rpm than peak power. The width of the torque plateau decides how relaxed a car feels.
- To pick a car, match the torque spread and power-to-weight to your driving. Do not chase the biggest number.
- Mileage = energy the car needs (weight, tyres, drag that grows with speed squared) ÷ how efficiently the engine and drivetrain deliver it.
- ARAI numbers compare cars; real-world city figures for the Creta variants run 17–46% lower.
- Small turbos help at cruise but cost fuel when you use the boost. Hybrids give the biggest real gain.
- BS6 made diesels costlier; E20 may trim km/l slightly for older petrols.

*Unverified or approximate: Swift kerb weight (925 kg) from a listing; power-to-weight bands, thermal-efficiency percentages, Skoda cylinder deactivation, BS6 phase dates and small-diesel withdrawal are textbook rules of thumb or general knowledge, not checked against an official document. Hybrid figures: Carlelo (ARAI 27.26 and 27.97 km/l). Creta ARAI claims and owner-reported mileage: Spinny's Creta mileage comparison (owner-reported, indicative only). Spec sources as in Part 1. Theory: Heywood, Internal Combustion Engine Fundamentals.

See also: Automobile Engineering 1/12 — The Engine, Part 1. Next: Part 3, Transmissions and drivetrain.

AUTOMOBILE ENGINEERING · PART 3 OF 12

Transmissions and Drivetrain

By Isagi · 9 November 2026 · All specs as of November 2026 (manufacturer brochures and listings)

TL;DR

The gearbox exists because an engine works well only inside a narrow band of revs, while a car must move from a standstill to highway speed. Gears trade speed for pulling force. In India you will meet five kinds of transmission: manual, AMT, torque-converter automatic, CVT and dual-clutch (DCT), plus the e-CVT in strong hybrids. They differ in cost, smoothness and efficiency. The **drivetrain layout** (front-, rear- or four-wheel drive) decides which wheels get the power. Most Indian cars are front-wheel drive; true off-roaders need four-wheel drive with a low-range gear.

1. Why a car needs gears

A petrol engine delivers useful power roughly between 1,500 and 6,000 rpm. Idle is far lower, and a car's wheels turn at a few hundred rpm at city speeds. The transmission bridges the gap. **Torque at the driven axle = engine torque × gear ratio × final-drive ratio**, before losses. Take an engine making 100 Nm, a first gear of 3.45, a final drive of 4.3 and a fifth gear of 0.77 (illustrative figures, not a specific car): first gear delivers about 1,480 Nm to the axle and fifth about 330 Nm. First gear gives a strong push for starting and climbing; fifth lets the engine turn slowly at highway speed, which saves fuel and noise.

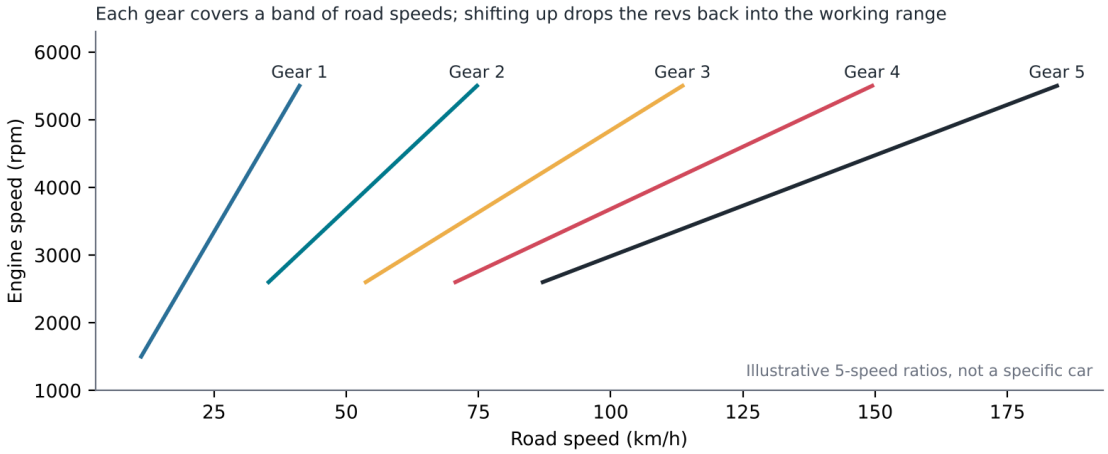


Fig 1. Each gear covers a band of road speeds. Shifting up drops the engine back to a lower rpm so it can keep accelerating.

More gears mean smaller steps between them, so the engine stays nearer its sweet spot. That is why 6-speed and 7-speed gearboxes help mileage compared with 4-speeds, and why the Part 2 point about wide torque plateaus matters: a wide plateau needs fewer shifts.

2. The manual: clutch and gears

A manual gearbox joins the engine to the wheels through a **clutch**, a pair of friction discs the driver presses apart or together. Pressing the pedal disconnects the engine so a gear can be chosen; releasing it reconnects. Manuals are cheap, light, efficient and durable, and they give the driver full control. Their weakness is the city: heavy traffic means constant clutch work. That is why automatic gearboxes now sell in large numbers in Indian metros.

3. The four kinds of automatic

All automatics take the clutch and gear selection away from the driver. They do it in four different ways, and the way decides how a car feels and what it costs to own.

Type	How it works	Strength	Weakness	Indian examples
AMT (automated manual)	A manual gearbox with a computer-controlled robot working the clutch and shifts	Cheapest automatic; manual-like mileage	Slow, jerky shifts, especially when pulling away	Maruti Swift AMT, Tata Punch AMT
Torque-converter automatic	A fluid coupling in place of the clutch, plus planetary gear sets	Smooth, tough, good for heavy loads and off-road	Heavier, costlier, a little less efficient	Creta diesel 6-speed automatic; Thar
CVT (continuously variable)	A belt running between two cone-shaped pulleys with no fixed gears	Very smooth; keeps revs in the best band	Rubbery feel when accelerating hard; less suited to high torque	Hyundai IVT (Creta petrol); Honda City
DCT (dual-clutch)	Two clutches, one for odd gears and one for even; the next gear is pre-selected	Fast shifts, sporty, efficient	Can be hesitant at crawling speed; costly to repair	Creta 1.5 turbo (7-speed DCT); Skoda 1.5 TSI (7-speed DSG)

The e-CVT in hybrids is a different idea: it uses a planetary gear set to blend power from the petrol engine and one or two electric motors, with no belt and no clutch. Toyota's Innova Hycross hybrid and the Hyryder use this layout, and it is why they feel smooth and need no shifts. Part 7 explains hybrids in full.

iMT (Hyundai and Kia's clutchless manual) is a halfway option: it has a manual gearbox with a sensor that handles the clutch, so you shift yourself but need no pedal.* Which type suits you depends on traffic, budget and how much you drive; Part 11 returns to that choice.

4. Drivetrain layouts: which wheels get the power

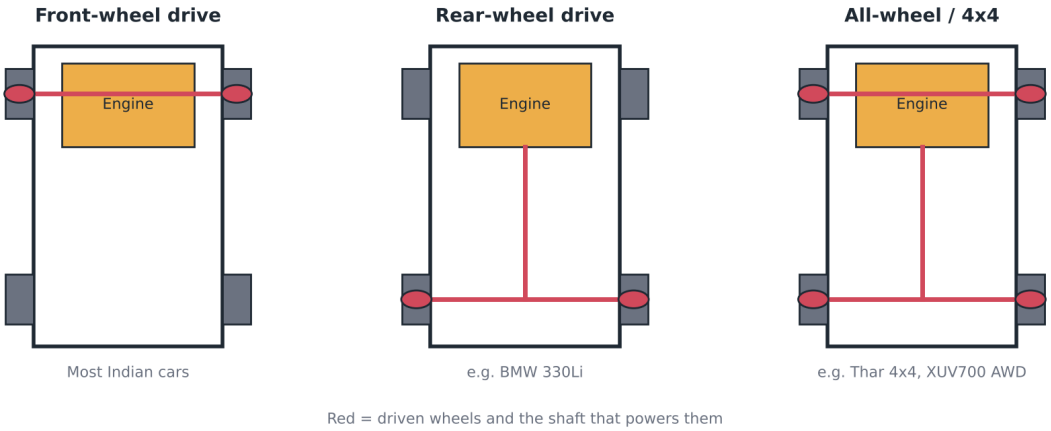


Fig 2. Three layouts. Red shows the driven wheels and the shaft that feeds them.

- **Front-wheel drive (FWD):** the engine and driven wheels sit together. It is cheap, compact and light, with good traction on wet roads. Almost every Swift, Nexon, Creta and City is FWD.
- **Rear-wheel drive (RWD):** a shaft carries power to the rear. It gives balanced weight and better handling at the limit, but it takes up floor space. The BMW 330Li is RWD; the Mahindra Thar offers RWD on some versions.*
- **All-wheel and four-wheel drive (AWD/4x4):** power goes to all four wheels. AWD systems (Mahindra XUV700 AWD*) work automatically on roads; a true 4x4 (Thar 4x4*) adds a *low-range* gearbox that multiplies torque again for steep, rough ground. Both add weight, cost and a fuel penalty.

A **differential** sits between the driven wheels so the outer wheel can turn faster than the inner one in a corner. An open differential sends power to whichever wheel slips first; *locking* differentials force both wheels to turn together, which is what off-roaders use on loose ground.

5. What it means for a buyer

- **City use, heavy traffic:** an automatic, preferably CVT, torque converter or DCT, is worth the premium; an AMT is the budget option.
- **Highway use:** a DCT or torque converter gives quick overtakes; a manual is cheaper and often a little more efficient.
- **Hills and rough roads:** torque-converter automatic or manual with strong low-end torque; for serious off-roading choose a 4x4 with low range.
- **Mileage:** manual ≥ AMT ≈ DCT ≥ CVT ≥ torque converter in most cases, but differences are small next to driving style (Part 2).*

Summary

- The transmission lets an engine with a narrow rev band move a car across all speeds; torque at the axle = engine torque × gear ratio × final drive.
- Manuals are cheap and efficient; automatics trade some of that for traffic comfort.
- AMT is a robot-driven manual: cheap but jerky. Torque converter is smooth and tough. CVT keeps revs steady. DCT is quick and sporty. e-CVT blends engine and motors in hybrids.
- FWD is common and cheap; RWD helps handling; AWD/4x4 helps traction and off-road ability, at a cost in weight and fuel.
- Choose by traffic, budget and terrain, not by the number of gears.

*Unverified or general knowledge: gear ratios in Section 1 are illustrative, not from a brochure; RWD/4x4 availability on the Thar and the XUV700 AWD, the iMT description and the mileage ordering in Section 5 are not checked against official documents. Transmission offerings confirmed in November 2026 listings: Swift 5-speed AMT; Creta petrol 6MT or IVT, turbo 7-speed DCT only, diesel 6MT or 6-speed automatic; Thar 6-speed torque-converter automatic; Honda City CVT; Hyryder e-CVT; Skoda 1.5 TSI 7-speed DSG only. Punch AMT and Innova Hycross e-CVT were not re-verified. Sources: Autocar India, Honda Cars India, Wikipedia. Theory: Heywood; Gillespie, Fundamentals of Vehicle Dynamics.

See also: the rest of the Automobile Engineering series (parts 1 to 12). Next: Part 4.

AUTOMOBILE ENGINEERING · PART 4 OF 12

Chassis, Suspension, Steering and Brakes

By Isagi · 9 November 2026 · All specs as of November 2026 (manufacturer brochures and listings)

TL;DR

The chassis is the skeleton that carries the engine, passengers and crash loads. Suspension keeps the tyres on the road over bumps, steering points the car, and brakes turn speed into heat. All four work through one small thing: the patch of rubber where each tyre meets the road. Stopping distance grows with the **square** of speed, which is why 120 km/h is far more dangerous than 80 km/h. For Indian roads the practical questions are ground clearance, ride comfort over potholes, and whether the car has ABS and ESC.

1. The chassis and body

Most modern cars use a **monocoque** (unibody): the floor, pillars and roof are one welded steel shell that is both body and frame. It is light, strong in a crash and cheap to mass-produce. Tough utility vehicles such as the Mahindra Thar and Scorpio-N use a separate **ladder frame**, two heavy steel rails with the body bolted on top.* It carries heavy loads and survives abuse, but is heavier and rides less smoothly. Electric cars add a new idea: the battery pack forms a flat structural floor (a 'skateboard'), which Part 6 covers.

2. Suspension: springs and dampers

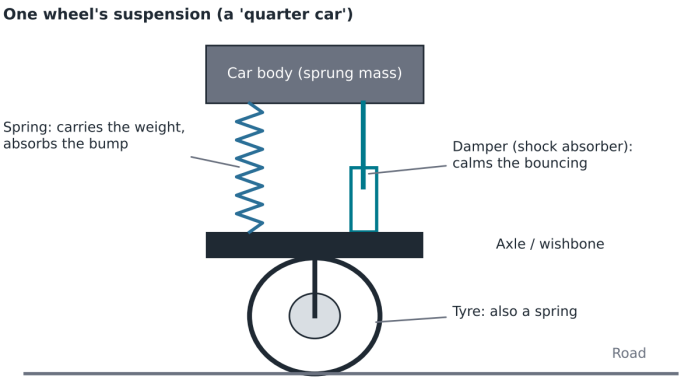


Fig 1. A spring carries the car's weight and absorbs bumps. A damper stops the spring from bouncing repeatedly. The tyre acts as a third, stiffer spring.

A spring alone would bounce up and down after every bump, so each wheel also has a **damper** (shock absorber) that converts the bouncing into heat. The art is the trade-off: soft springs ride comfortably but let the body lean in corners; stiff springs corner flat but thump over potholes. Indian buyers usually want comfort and high ground clearance, so most cars are set softer and taller than European equivalents.*

- **McPherson strut (front):** a strut combining spring and damper on a single bolt-on unit. It is compact and cheap, and used on the front of almost every mass-market car.*
- **Torsion beam (rear):** two trailing arms joined by a twisting cross-beam. Simple and cheap; common on hatchbacks and compact SUVs.*
- **Multi-link (rear):** several arms control each wheel separately, giving better ride and handling at higher cost. Typical in premium cars.

- **Solid axle with leaf or coil springs:** a rigid beam across the rear, tough for loads and off-road. Typical in ladder-frame SUVs and pickups.*

Ground clearance is the gap between the lowest part of the underside and the road. India's speed breakers and broken roads make it a headline spec for SUV buyers. Higher clearance helps on bad roads but raises the centre of gravity, which increases body roll.

3. Steering

Almost all cars use a **rack and pinion**: a small gear (pinion) on the steering column moves a toothed bar (rack) sideways, turning the front wheels. A power-assist system lightens the effort. Older cars used hydraulic pumps driven by the engine; modern ones use **electric power steering (EPS)**, a motor on the steering column. EPS saves fuel (no pump drag) and can vary the weight with speed, so the wheel is light in parking and firmer on the highway. The spec sheet's **turning radius** tells you how tight a U-turn the car can make, which matters more in a crowded city than 0–100 km/h time.

4. Brakes

Brakes work by friction: pads clamp a disc (or shoes press a drum) and convert the car's kinetic energy into heat. Kinetic energy is ½ × mass × speed², so energy, and therefore braking distance, grows with the square of speed. Add the distance covered while the driver reacts (about 1.5 seconds is a common assumption) and the numbers rise fast.

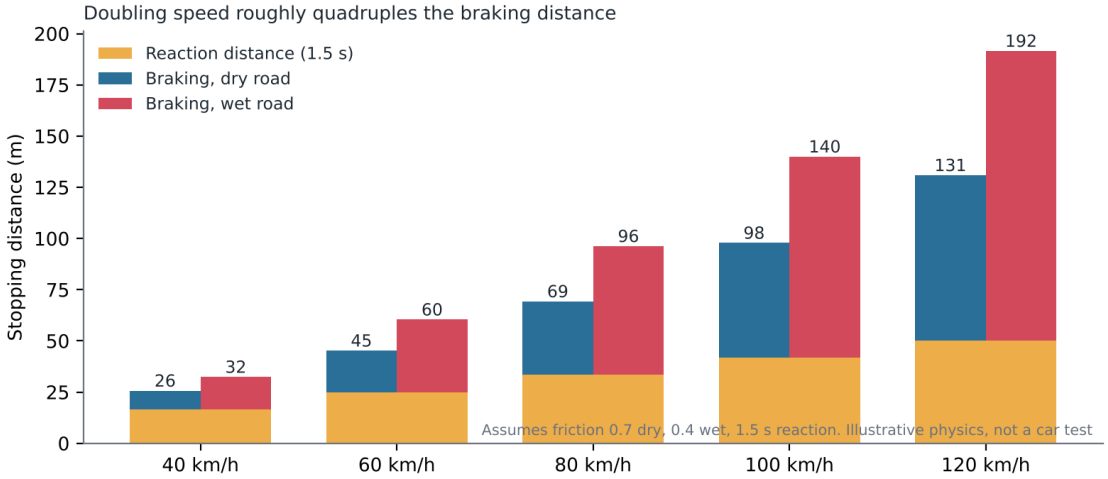


Fig 2. Total stopping distance on dry and wet roads. At 120 km/h a typical car needs roughly 130 m on a dry road and well over 150 m when wet, using simple physics with the assumptions shown.

- **Disc vs drum:** discs shed heat better and resist fade; drums are cheaper and still used on the rear wheels of many small cars.*
- **ABS (anti-lock braking):** pulses the brakes many times a second so the wheels never lock, letting you steer while braking hard. It has been compulsory on new cars in India for several years.*
- **EBD and brake assist:** distribute braking force between front and rear, and add full force when the system senses a panic stop.
- **ESC (electronic stability control):** senses a skid and brakes individual wheels to straighten the car. Many safety experts rate it as important as airbags; Part 5 covers it.
- **Regenerative braking (EVs, hybrids):** the motor works as a generator when you lift off, returning some energy to the battery. It cuts brake wear and boosts efficiency in the city.

5. Tyres: the part that touches the road

Each tyre's contact patch is only about the size of a palm, and that patch carries every force: acceleration, braking and cornering.* A size such as **215/60 R17** reads: 215 mm tread width; sidewall height equal to 60% of that width (about 129 mm); R for radial construction; 17-inch wheel. A taller sidewall (higher aspect ratio) cushions bumps; large wheels with thin sidewalls look sporty but ride harder and are costlier to replace. Tyre pressure matters for safety, comfort and mileage (Part 2), and is the cheapest thing a driver can check.

6. What it means for a buyer

- Look at ground clearance and the rear suspension type if you drive on poor roads.
- Test the ride over a speed breaker at 20–30 km/h; that tells more than a spec sheet.
- Insist on ABS and ESC, particularly on faster cars and for highway use.
- Check turning radius and steering lightness if most of your driving is in the city.

Summary

- A monocoque is a light unibody; a ladder frame is heavy and tough for utility vehicles.
- Springs carry weight; dampers control bouncing; stiffer means sportier, softer means more comfortable.
- Rack-and-pinion steering with electric assist is standard; turning radius matters in cities.
- Brakes turn kinetic energy (½ mv^2) into heat; stopping distance rises with the square of speed.
- ABS stops wheels locking; ESC stops skids; regenerative braking saves energy in EVs and hybrids.
- A tyre size code tells width, sidewall height and wheel diameter.

*Unverified or general knowledge: which cars use ladder frames, which suspension types are common, the 'palm-sized' contact patch, drum-brake use on rear wheels, the timing of the ABS mandate and the claim that Indian cars are set softer than European ones are not checked against official documents. Stopping distances in Fig 2 are calculated from simple physics with stated assumptions (friction 0.7 dry, 0.4 wet, 1.5 s reaction); real cars and tyres vary. Theory: Gillespie, Fundamentals of Vehicle Dynamics.

See also: the rest of the Automobile Engineering series (parts 1 to 12). Next: Part 5.

AUTOMOBILE ENGINEERING · PART 5 OF 12

Safety Engineering

By Isagi · 9 November 2026 · All specs as of November 2026 (manufacturer brochures and listings)

TL;DR

A car's safety has three jobs: help the driver **avoid** a crash, help everyone **survive** one, and help rescuers afterwards. Avoidance comes from brakes, tyres, ABS and stability control. Survival comes from a rigid passenger cell, crumple zones that absorb energy, seatbelts and airbags. In India, crash ratings from Bharat NCAP (government, since 2023) and Global NCAP (independent) now let buyers compare cars on tested performance instead of marketing. A star rating is meaningful only for the exact version tested and the protocol year it was tested under.

1. The physics: why speed is the enemy

In a crash, the car's kinetic energy (½ × mass × speed²) has to go somewhere. It goes into bending metal, which is the safety engineer's job to arrange. Doubling the speed quadruples the energy. Bharat NCAP's frontal offset test runs at 64 km/h; a crash at 120 km/h carries about 3.5 times that energy.* No structure can absorb that safely, which is why speed limits matter more than any feature.

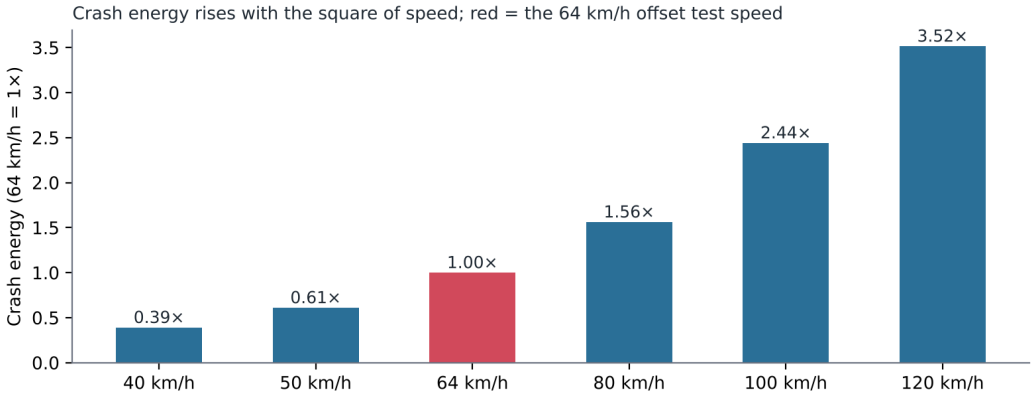


Fig 1. Relative crash energy against the 64 km/h test speed. Energy = ½ mv², so it scales with speed squared.

2. Three layers of safety

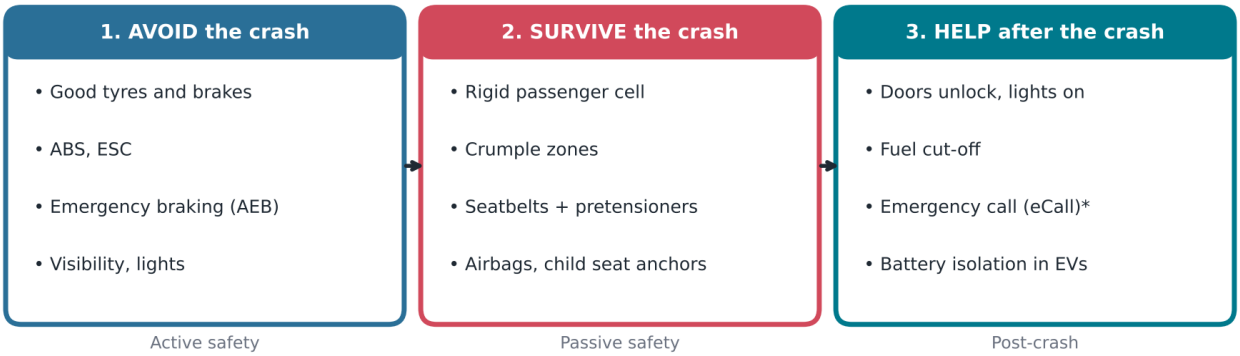


Fig 2. Safety as three layers: avoid, survive, help after the crash. (*eCall is not yet widely fitted in India.)

3. How a car survives a crash

- **Rigid passenger cell:** the cabin is built from very strong steel (often high-strength or hot-stamped grades) so it does not collapse onto occupants.
- **Crumple zones:** the front and rear are designed to fold progressively, absorbing energy over a longer time. Stopping over a longer time means lower force on the body.
- **Seatbelts with pretensioners and load limiters:** the belt tightens in the first milliseconds and then lets out a little so the chest is not injured. The belt is the single most important restraint; airbags assume you are wearing it.
- **Airbags:** a front airbag cushions the head and chest; side and curtain airbags protect in side impacts and rollovers. Six airbags are not mandatory in India: a January 2022 draft notification lapsed, and in September 2023 the transport minister said six airbags would not be made compulsory and that Bharat NCAP should push makers to fit them. Several popular models, including the Swift and Dzire, now fit six as standard.
- **Child restraints (ISOFIX):** standard anchor points let a child seat clip rigidly into the car. Child occupant protection is scored separately in NCAP tests.

4. How a car avoids a crash

- **ABS** prevents wheel lock under hard braking; **ESC** stops skids by braking single wheels. Global NCAP has pushed ESC as a key feature, and it is now rewarded in the protocol (see below).
- **Autonomous emergency braking (AEB)** uses a radar or camera to brake automatically when a collision is near. It is part of the driver-assistance systems in Part 8.
- **Tyres and brakes in good condition** matter more than any electronic feature; Part 4 covers stopping distances.

5. Crash tests: how to read the stars

Global NCAP, a UK-based charity, started crash-testing popular Indian cars in 2014 under its Safer Cars for India campaign. Its first round tested five cars, including the Swift, with no frontal airbags fitted; all scored zero stars for adult protection. Results pushed Toyota and Volkswagen to make frontal airbags standard across their Indian lineups.

Bharat NCAP is India's own programme. Its programme took effect on 1 October 2023 (announced 22 August 2023*), and crash tests began in December 2023. The current protocol (AIS-197) uses a 64 km/h frontal offset test, a 50 km/h side-barrier test and a pole side test, and carries ESC and pedestrian-protection requirements. The first 5-star results went to the Tata Harrier and Safari (adult 30.08/32, child 44.54/49). A tougher Bharat NCAP 2.0 (five crash tests, a 100-point scale, with curtain airbags and ESC required for any star rating) is in draft for October 2027. The programme is voluntary: makers choose which cars to submit.

Car and test	Protocol / year	Result
Maruti Dzire (4th gen)	Global NCAP, Nov 2024	Adult 31.24/34 (5 stars); child 39.20/49 (4 stars); first Maruti to score 5 stars
Maruti Dzire	Bharat NCAP	5 stars (29.46/41.57)
Maruti Swift (3rd gen)	Global NCAP 2.0, 2022	Adult 19.19/34, child 16.68/49 (reported as 1 star each; not confirmed)
Tata Punch (petrol)	Global NCAP, 2021 (older 17-point scale)	Adult 16.45/17 (5 stars); child 40.89/49 (4 stars). Not comparable with 34-point scores
Tata Punch.ev	Bharat NCAP, 2024	Adult 31.46/32 (5 stars); child 45.00/49 (5 stars); the 2026 facelift scored 31.09/32 and 45/49

Car and test	Protocol / year	Result
Tata Nexon	Global NCAP and Bharat NCAP	5 stars in both (Bharat NCAP 29.41/43.83)
Tata Harrier, Safari	Bharat NCAP, Dec 2023	First Bharat NCAP results: adult 30.08/32, child 44.54/49 (5 stars each)
VW Taigun, Skoda Kushaq	Global NCAP, Oct 2022	Adult 29.64/34, child 42.00/49 (5 stars each)
Hyundai Creta	Global NCAP, 2022 (older protocol)	3 stars (reported)

Not every popular car has a Bharat NCAP rating yet: as of November 2026 the Swift, Creta, Hyryder and City have no confirmed Bharat NCAP result (the Kushaq and Slavia have Global NCAP ratings only).

Why scores aren't directly comparable. Global NCAP changed its protocol in July 2022 (adding side-impact protection and rear seatbelt reminders, and making ESC and side head airbags requirements for higher ratings) and again in August 2025 (rewarding safety equipment only if fitted as standard). A 5-star result in 2021 and one in 2024 reflect different tests. Always check which protocol, which variant (airbag count differs by trim) and which year.

6. What it means for a buyer

- Choose the variant that matches the tested one: the rating usually applies to a specific trim and airbag count.
- Check which protocol and year the rating came from; newer protocols are stricter.
- Prefer cars with six airbags, ESC and ISOFIX as standard, not as paid extras.
- Wear the belt in the rear seat too; most rear passengers in India still do not.

Summary

- Safety has three layers: avoid the crash, survive it, help afterwards.
- Crash energy rises with speed squared, so a structure that is safe at 64 km/h cannot protect at 120 km/h.
- A rigid cell, crumple zones, belts with pretensioners and airbags protect occupants together.
- ABS, ESC and AEB help avoid the crash.
- Bharat NCAP (since 2023) and Global NCAP rate cars; ratings depend on the variant, the protocol and the year.
- Prefer six airbags, ESC and ISOFIX as standard.

*Unverified or general knowledge: the 3.5× comparison is arithmetic ($(120/64)^2$); the 22 August 2023 Bharat NCAP announcement date, eCall availability and the seatbelt-usage remark were not re-checked, and the Swift's star ratings (reported as 1 star adult and 1 star child) and the Creta result were not confirmed from Global NCAP. Cross-checked in November 2026: six-airbag status (MoRTH draft of 14 January 2022; minister's statement of 13 September 2023), Bharat NCAP dates, protocol and results, Dzire, Punch, Punch.ev, Nexon, Harrier, Safari, Kushaq results (Autocar India, Autocar Professional, CarDekho, Team-BHP and others). Protocol history: Global NCAP (via Wikipedia).

See also: the rest of the Automobile Engineering series (parts 1 to 12). Next: Part 6.

AUTOMOBILE ENGINEERING · PART 6 OF 12

Electric Vehicles

By Isagi · 9 November 2026 · All specs as of November 2026 (manufacturer brochures and listings)

TL;DR

An electric car swaps the engine, fuel tank and gearbox for a battery, an inverter and a motor. The battery's size in **kWh** is its fuel tank; the motor's torque is available from the first turn of the wheels. Running costs per km are far lower than petrol or diesel, but the car costs more to buy, and what matters in practice is whether you can charge at home. Claimed range is a lab figure: the Nexon EV's 489 km claim compares with a user-reported 297 km. GST on EVs is 5%, against 18% or 40% for most petrol and diesel cars.

1. How an electric car works

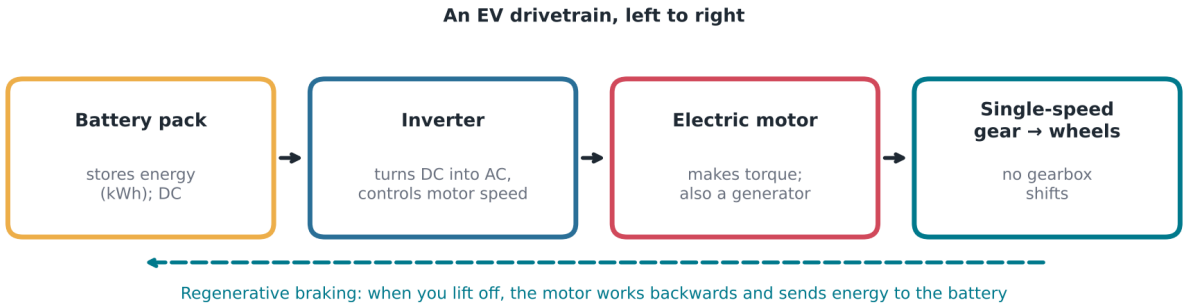


Fig 1. The EV drivetrain: battery, inverter, motor and a single-speed gear. Regeneration sends energy back to the battery.

- **Battery pack:** thousands of cells storing energy as direct current. Capacity is measured in kilowatt-hours (kWh): a 45 kWh pack can deliver 45 kW for one hour.
- **Inverter:** converts the battery's DC into AC for the motor and controls its speed and torque.
- **Motor:** an electric motor makes its peak torque from standstill, with no need to rev up. The Nexon EV (45 kWh) is listed at 145 hp and 215 Nm; the Mahindra BE 6 (2024 launch version, 79 kWh pack) at 210 kW (282 hp) and 380 Nm. This is why EVs feel quick off the line.
- **Single-speed gear:** electric motors work over a huge rev range, so one fixed gear is enough. There are no shifts.
- **Regeneration:** lifting off the accelerator turns the motor into a generator, recovering part of the energy that a petrol car loses as brake heat.

2. The battery: the costly heart

The battery is the most expensive part of an EV, commonly estimated at 30–40% of the car's cost.* Two chemistries dominate: **LFP** (lithium iron phosphate), cheaper and longer-lived, and **NMC** (nickel-manganese-cobalt), more energy-dense and costlier.* A battery slowly loses capacity over years and charge cycles, which affects resale value; makers typically offer an eight-year battery warranty.* A thermal-management system heats or cools the pack, because batteries age faster when hot.

Importantly, EV platforms put the flat battery in the floor like a skateboard. That lowers the centre of gravity and frees cabin space. The Mahindra BE 6 and XEV 9e sit on Mahindra's INGLO platform; the Tata Punch.ev is the first car on Tata's acti.ev architecture.

3. Charging

Method	Typical power	What it takes (Nexon EV 45 kWh)
Ordinary 15 A home socket	about 3 kW	about 17.6 hours for 10–100%
Home AC wall box	7.2 kW	about 6.6 hours for 10–100%
Public DC fast charger	60 kW	about 40 minutes for 10–80%
Fast charging on newer platforms	140–180 kW	BE 6 Sporteq: under 20 minutes for 20–80% (59 kWh at 140 kW, 70 kWh at 160 kW, 79 kWh at 180 kW)

A rule of thumb: charging time ≈ energy needed ÷ charger power, plus extra at the top because the car slows charging above about 80%. Fast chargers are best for highway trips; overnight home charging does the daily work. If you cannot charge at home or work, an EV is much less convenient.

4. Range: claimed versus real

Claimed range comes from a standard lab test. Real range depends on speed (drag grows with the square, as in Part 2), air conditioning, temperature, load and driving style. The Nexon EV 45 kWh is listed at a claimed 489 km, while one listing gives a user-reported 297 km, about 39% lower. The BE 6 Sporteq (relaunched 15 August 2026) claims 548, 651 and 683 km for its 59, 70 and 79 kWh packs. As a quick check, divide the pack size by the range to get consumption: the Nexon's real-world figure works out to about 15 kWh per 100 km, against about 9 kWh per 100 km on the claim.

5. The money question

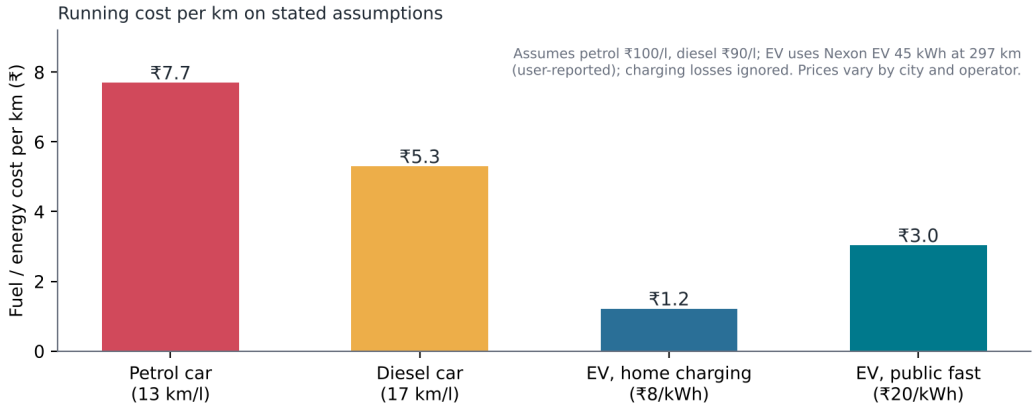


Fig 2. Energy cost per km for a petrol car, a diesel car, and an EV charged at home or at a public fast charger, on the assumptions shown.

On these assumptions, home-charged EV running costs are roughly one-sixth of petrol. Public fast charging is still well under half. But an EV costs more upfront, so the saving only pays back with enough kilometres. A simple break-even: **extra purchase price ÷ saving per km**. If an EV costs ₹4 lakh more and saves ₹6 per km, you need about 67,000 km to recover it (about 5–6 years at 12,000 km a year).

- **Tax:** EVs pay 5% GST; since 22 September 2025 most petrol and diesel cars pay 18% (small cars) or 40% (larger cars). This narrows the price gap. There is no central purchase subsidy for private electric cars under PM E-DRIVE (its electric two-wheeler support was extended to 31 March 2028), so the 5% GST is the main tax benefit.
- **Unknowns:** battery life, resale value and charging availability are still uncertain in India.
- **Best fit:** a household with home charging, mostly city and short intercity use, ideally as a second car.

Summary

- An EV = battery (kWh) + inverter + motor + a single-speed gear; torque is instant.
- The battery is the costly part (roughly 30–40% of the car's cost) and its chemistry and warranty matter.
- Charging time ≈ energy ÷ charger power; home AC does the daily work, DC fast chargers handle trips.
- Claimed range is a lab figure; the Nexon EV example shows real range about 40% below the claim.
- Running cost per km is far below petrol, but the purchase premium means the saving needs enough km to pay back.
- GST is 5% on EVs versus 18% or 40% on most other new cars.

*Unverified or general knowledge: battery cost share (30–40%), LFP vs NMC description and the 8-year warranty norm were not checked against official sources. Cross-checked in November 2026: GST rates (56th GST Council), Nexon EV (145 hp, 215 Nm, 489 km claimed, 10–100% in 6 h 36 min on 7.2 kW; Autocar India, CarDekho), BE 6 Sporteq (59/70/79 kWh, 548/651/683 km, price ₹20.20–27.70 lakh ex-showroom; Autocar India), PM E-DRIVE (Ministry of Heavy Industries notification, 10 August 2026). The earlier BE 6 power and torque figures (210 kW, 380 Nm) are from its 2024 launch and were not re-verified for the Sporteq. Fuel prices in Fig 2 are assumptions, not quotes. Charging losses (about 10%) are ignored. Break-even example uses made-up figures to show the method.

See also: the rest of the Automobile Engineering series (parts 1 to 12). Next: Part 7.

AUTOMOBILE ENGINEERING · PART 7 OF 12

Hybrids, CNG and Ethanol

By Isagi · 9 November 2026 · All specs as of November 2026 (manufacturer brochures and listings)

TL;DR

Between a pure petrol car and a pure EV sit three alternatives. **Hybrids** add an electric motor and a small battery to a petrol engine, giving big efficiency gains in city traffic without any charging. **CNG** burns compressed natural gas: low fuel cost, with some loss of power and boot space. **Ethanol** blended into petrol (E20 today) is mostly a national energy policy that affects your car slightly. On claimed figures the Honda City hybrid and the Toyota Hyryder hybrid return about 27 to 28 km/l, about 57% to 61% more than the 17.4 km/l petrol Creta, although these are different cars.

1. The hybrid family

Type	What it adds	Can it drive on electric alone?	Examples
Mild hybrid	Small 48-volt motor assisting the engine	No	Mercedes C-Class (48V, up to 20 hp and 200 Nm boost)
Strong hybrid	Bigger motor and battery, can move the car alone	Yes, at low speed	Toyota Hyryder, Innova Hycross, Honda City e:HEV
Plug-in hybrid (PHEV)	Large battery you can charge from a socket	Yes, for tens of km	MG Hector PHEV*

Strong hybrids matter most to mass-market buyers because they need no plug. Their trick is to let the electric motor do the work the engine does worst: pulling away and crawling in traffic. The petrol engine then runs mostly at steady, efficient loads, and the motor recaptures braking energy that would otherwise be lost.

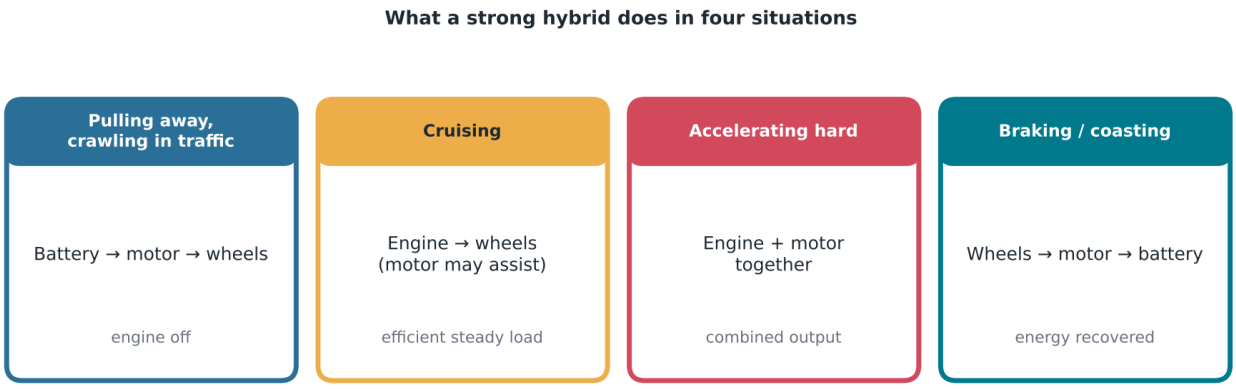


Fig 1. What a strong hybrid does in four common situations.

Both hybrids on sale in the 1.5-litre class use the Atkinson cycle, a variant in which the intake valve stays open longer so the engine uses less of the air it takes in each cycle, trading peak power for efficiency; the electric motor makes up the missing punch.* The numbers show how the two makers split the work (in PS, the metric horsepower used in brochures; 1 PS is about 0.986 bhp):

	Toyota Hyryder hybrid	Honda City e:HEV
Engine	1.5 L, 3-cylinder	1.5 L, 4-cylinder (Atkinson cycle)

	Toyota Hyryder hybrid	Honda City e:HEV
Engine power / torque	92.45 PS / 122 Nm	98 PS / 127 Nm
Electric motor power / torque	80 PS / 114 Nm	109 PS / 253 Nm
Combined power	115.56 PS (about 114 bhp)	126 PS (about 124 bhp)
ARAI claimed efficiency	27.97 km/l	27.26 km/l

The City's motor is stronger than its engine and does most of the driving in many conditions; Toyota's system is slightly more efficient on paper. The Innova Hycross hybrid uses a 2.0-litre engine plus motor for about 184 bhp of system output, so the same idea scales up to a family MPV (Part 1).

2. CNG: cheap fuel, small penalties

CNG is natural gas stored under high pressure in a steel cylinder. The engine runs on it with a modified fuel system and usually can switch back to petrol. The Maruti Swift S-CNG shows both the benefit and the cost:

- **Efficiency:** 32.85 km/kg claimed (ARAI).
- **Power:** about 69.75 PS (68.8 bhp) and 101.8 Nm, against 80.46 bhp and 111.7 Nm for the petrol engine, which is roughly 14% less power and 9% less torque.
- **Boot space:** the gas cylinder usually takes up luggage room; some makers fit twin smaller cylinders to keep more boot (for example Tata's iCNG range*).
- **Availability:** CNG stations exist mainly in some cities and along certain corridors, so check your area before buying.

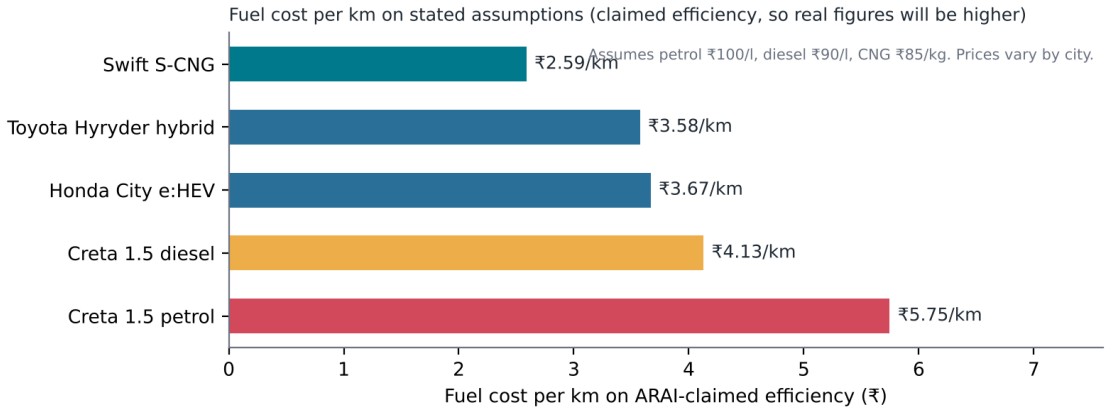


Fig 2. Fuel cost per km using claimed efficiency and the assumed prices shown. Different cars, so read the gaps as scale, not an exact comparison.

On these assumptions the CNG Swift costs under half as much per km as the petrol Creta. These are different cars, so compare like for like within one model before deciding; still, the gap is why CNG dominates taxi fleets and high-mileage family cars. The break-even logic of Part 6 applies again: divide the CNG price premium by the saving per km to see how many kilometres it takes to pay back.

3. Ethanol: E20 and flex fuel

Ethanol is alcohol made from sugarcane or grain. Petrol in India is now sold blended with 20% ethanol (**E20**), which is the standard pump fuel nationwide. Ethanol holds less energy per litre than petrol (roughly two-thirds), so E20 has about 7% less energy per litre than pure petrol, by this arithmetic: 0.8 × 100 + 0.2 × 66 ≈ 93.* India's petroleum ministry (August 2025) put the mileage loss at about 1–2% for E10-designed cars calibrated for E20 and 3–6% for others, and the transport minister told Parliament in July 2026 that the drop is 2–6%. Newer engines are calibrated for E20 (the Swift is listed as E20-compatible). Older cars may need compatible rubber hoses and seals over time.

Flex-fuel engines can run on any blend up to nearly pure ethanol. They are a policy bet to cut crude-oil imports and support farmers. Critics raise concerns about food-versus-fuel land use, water use and lower mileage; supporters point to import savings and lower tailpipe carbon.* Part 10 looks at how these policies shape the market.

4. Which one for whom

- **Mostly city driving, wants an automatic and low fuel bills with no charging:** strong hybrid.
- **High annual kilometres on a tight budget, with a CNG station nearby:** CNG.
- **Can charge at home and drives short distances:** EV (Part 6).
- **Long highway runs, heavy loads:** diesel or a plain efficient petrol still make sense (Part 2).

Summary

- Mild hybrids assist the engine; strong hybrids can drive on electric alone; plug-ins add a chargeable battery.
- A strong hybrid shines in city traffic because the motor handles low-speed work and recovers braking energy.
- The City e:HEV and Hyryder hybrid claim about 27–28 km/l, compared with 17.4 km/l for a petrol Creta.
- CNG cuts fuel cost per km sharply (the Swift S-CNG costs under half the petrol Creta per km on claimed figures, though they are different cars) but costs about 14% of power and some boot space on the Swift.
- E20 petrol carries about 7% less energy per litre; official estimates of mileage loss are 1–6%.
- Match the fuel to your kilometres, traffic and access to CNG or charging, not to the headline mileage.

*Unverified or general knowledge: PHEV example (MG Hector), Atkinson-cycle description, Tata iCNG twin-cylinder layout, E27/E30 plans, ethanol energy content (about two-thirds of petrol, textbook value) and the critics-and-supporters summary of flex-fuel policy were not checked against official documents. Cross-checked in November 2026: Hyryder hybrid (92.45 PS engine, 80 PS motor, 115.56 PS combined, 27.97 km/l; Autocar India), City e:HEV (98 PS engine, 109 PS motor, 126 PS combined, 27.26 km/l; Honda Cars India), Swift S-CNG (32.85 km/kg, 69.75 PS, 101.8 Nm; Team-BHP, Spinny), E20 mileage statements (Petroleum Ministry August 2025; Parliament reply July 2026, via Tribune, DriveSpark). Fuel prices in Fig 2 are assumptions, not quotes. Innova Hycross hybrid output was not re-verified.

See also: the rest of the Automobile Engineering series (parts 1 to 12). Next: Part 8.

AUTOMOBILE ENGINEERING · PART 8 OF 12

Software, Connectivity and Driver Assistance

By Isagi · 9 November 2026 · All specs as of November 2026 (manufacturer brochures and listings)

TL;DR

A modern car is a computer network on wheels. Dozens of small computers (ECUs) run the engine, brakes, airbags and screens, and talk to each other over an internal network. Newer cars add internet connectivity for apps, remote functions and over-the-air (OTA) updates, and cameras and radars for **ADAS** (advanced driver assistance systems). In India the ADAS you can buy today is Level 2 on the international scale: the car can help with steering and speed, but **you** remain responsible. ADAS has reached mainstream prices (the Mahindra XUV 3XO offered Level 2 ADAS from ₹11.99 lakh, in its AX5 L variant, at its April 2024 launch), but it was designed for orderly roads, and Indian roads are not always orderly.

1. The car as a network of computers

Each function has its own small computer, an **electronic control unit (ECU)**: engine management, transmission, ABS and ESC, airbags, air conditioning, instrument cluster, infotainment. A premium car can have well over a hundred.* They share data over an in-vehicle network, usually the **CAN bus**, so that for example the wheel-speed signal used by ABS is also available to the stability controller and the speedometer. The diagnostic port under the dashboard (OBD) lets a mechanic read fault codes from these ECUs, and BS6 norms made on-board diagnostics stricter (Part 2).

The industry buzzword is the **software-defined vehicle**: features that were once fixed in hardware are controlled by software that can be updated, and in some cases unlocked later through a paid subscription. That changes the business model: a car can earn revenue after the sale, and it also means two cars with identical hardware can differ in what they let you use.

2. Connectivity: from screens to over-the-air updates

- **Infotainment:** touchscreen, Android Auto and Apple CarPlay, voice assistants and navigation. Screen size and speed of response are now top reasons people choose one variant over another.
- **Connected-car apps:** a built-in SIM card links the car to a phone app for remote lock and unlock, location, climate pre-cooling and trip data. Most major brands offer one (Hyundai Bluelink, Kia Connect, MG iSMART, Tata iRA, Mahindra Adrenox, Maruti Suzuki Connect).*
- **OTA updates:** like a phone, a car can download new software and fix bugs without a visit to the workshop. Some updates are minor (maps, screen layouts); others change how the car drives or charges.

The trade-offs: a connected car collects data about where you go and how you drive. India's Digital Personal Data Protection Act, 2023 applies to personal data collected this way, although the rules for vehicles specifically are still evolving.* Anything online can in principle be attacked, so makers must secure the network. And long-term support matters: a car sold today must still receive updates in eight years.

3. ADAS: how far automation goes

Engineers describe how much of the driving a car does with a six-level scale from SAE International, from Level 0 (nothing) to Level 5 (everything, everywhere).

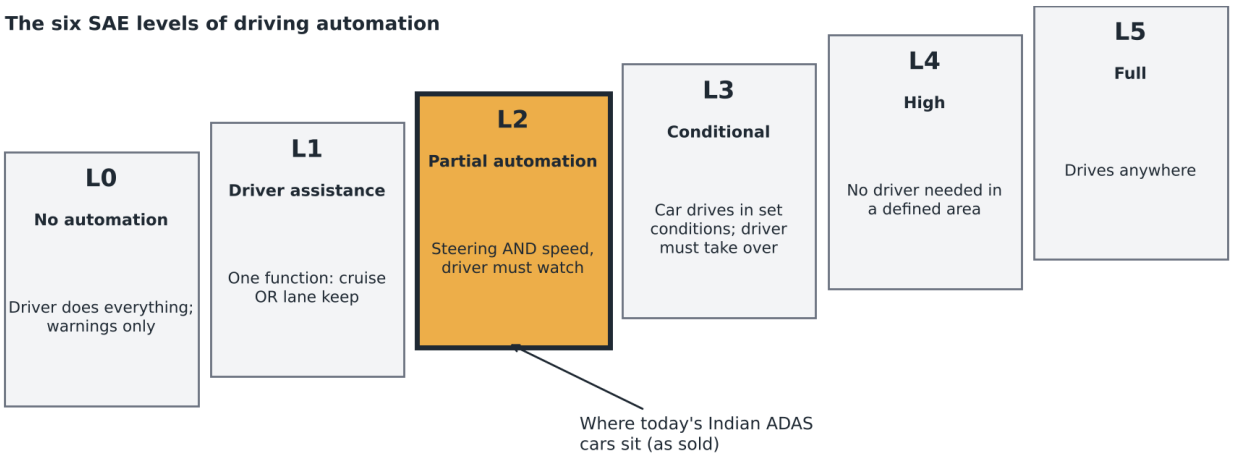


Fig 1. The six levels of driving automation defined by SAE International. Cars sold in India with ADAS are Level 2.

At **Level 2**, the car can steer and control speed together, for example adaptive cruise control with lane centring. But the driver must keep watching and be ready to take over at once. Levels 3 to 5 would let the car take responsibility in some or all conditions; they are not yet part of ordinary Indian car sales.*

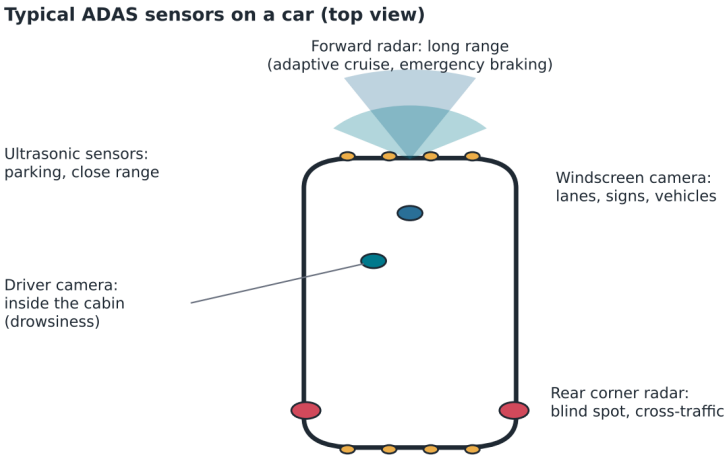


Fig 2. Where ADAS sensors typically sit. A forward camera and radar do most of the work; ultrasonic sensors handle parking; rear corner radars watch blind spots.

- **Autonomous emergency braking (AEB):** brakes if a collision is imminent. It is the most valuable feature, as Part 5 notes.
- **Adaptive cruise control (ACC):** keeps a set distance from the car in front, slowing and speeding up with traffic.
- **Lane departure warning and lane keep assist:** alert you or nudge the steering if you drift.
- **Blind-spot monitoring and rear cross-traffic alert:** warn about cars you cannot see.
- **Driver monitoring:** a cabin camera detects drowsiness or distraction.

4. ADAS meets Indian roads

ADAS was largely developed on roads with clear lane markings, predictable traffic and few unexpected obstacles. In India, markings are often faded or missing, two-wheelers filter through gaps, pedestrians and animals appear suddenly, and construction zones change the road layout. Lane-keeping depends on seeing markings, and emergency braking can trigger falsely or miss an unusual obstacle. Makers tune systems for local conditions, but owners often report that features behave inconsistently, so expect assistance, not autopilot.*

5. A buyer's checklist

- Check **which** ADAS functions are included and in which variant; the same name can cover very different feature sets.
- Test them in real traffic on a test drive, not only on an empty road.
- Ask how long software updates are supported and whether any features need a subscription.
- Read the data and privacy terms of the connected-car app.
- Treat ADAS as a safety net, not a replacement for attention.

Summary

- A car is a network of ECUs sharing data over a CAN bus; software increasingly defines what the car can do.
- Connectivity brings apps, remote features and OTA updates, and raises questions about data, security and long-term support.
- ADAS on sale in India is Level 2: the car assists, the driver stays responsible.
- AEB is the most valuable ADAS feature; others add comfort.
- Indian road conditions challenge systems designed for orderly roads; test them in real traffic.

*Unverified or general knowledge: the number of ECUs in premium cars, the list of connected-car brands, the status of data-protection rules for vehicles, the absence of Level 3+ cars in ordinary Indian sale and the statements on how ADAS behaves on Indian roads are general knowledge or reported owner experience, not tested here. Cross-checked in November 2026: the XUV 3XO launched on 29 April 2024 at an introductory ₹7.49 lakh for the base MX1 variant, which has no ADAS; the cheapest ADAS variant at launch was the AX5 L at ₹11.99 lakh (Autocar India, Mahindra press release); current range ₹7.79–15.04 lakh. SAE levels: SAE International standard J3016.

See also: the rest of the Automobile Engineering series (parts 1 to 12). Next: Part 9.

AUTOMOBILE ENGINEERING · PART 9 OF 12

Platforms, Manufacturing and Supply Chain, and How to Read a Spec Sheet

By Isagi · 9 November 2026 · All specs as of November 2026 (manufacturer brochures and listings)

TL;DR

A car company's biggest costs are fixed: developing the car and building the plant. So makers share a **platform** (the floor, suspension mounts and electrical architecture) across many models, so that one development bill is spread over lakhs of cars. Inside a plant, steel becomes a body, the body is painted, the powertrain is fitted and the car is inspected. Behind the plant sits a supply chain of thousands of parts suppliers, which is where disruptions (chips, rare-earth magnets, battery cells) hit. The chapter ends with a practical skill: how to read a spec sheet and turn its numbers into meaning.

1. Platforms: one design, many cars

A platform is the shared underbody and its engineering: wheelbase options, mounting points for engine and suspension, crash structure and wiring. Cars on one platform can look and feel quite different (hatchback, sedan, SUV) while sharing many unseen parts. The economics are the point: if developing a platform costs ₹2,000 crore (an illustrative number) and it underpins 5 lakh cars, development adds ₹40,000 to each car; on 1 lakh cars it adds ₹2 lakh. **Break-even volume = fixed costs ÷ (price − variable cost per car)**, so volume is what makes a car profitable.

Platform	Maker	Cars built on it (examples)
Heartect	Maruti Suzuki	Swift, Dzire
ALFA-ARC	Tata Motors	Punch, Altroz (petrol)
acti.ev	Tata Motors	Punch.ev (first car on it)
INGLO	Mahindra	BE 6, XEV 9e (electric, rear-motor layout)
MQB-A0-IN	Volkswagen Group (Skoda, VW)	Kushaq, Slavia, Taigun, Virtus
TNGA GA-C*	Toyota	Innova Hycross

Notice the electric platforms: Tata and Mahindra built dedicated EV platforms (acti.ev, INGLO) rather than modifying petrol cars. A purpose-built floor lets the battery sit flat under the cabin, which improves space and handling (Part 6).

2. Inside a car plant

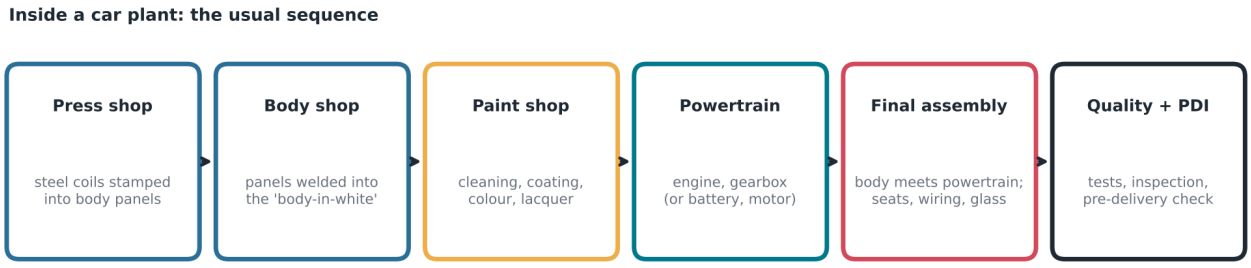


Fig 1. A typical assembly sequence. The 'body-in-white' is the bare welded shell before paint and parts.

- **Press shop:** giant presses stamp steel coils into doors, roofs and floors.

- **Body shop:** robots weld hundreds of panels into the body-in-white. Weld quality decides crash strength.
- **Paint shop:** the body is cleaned, rust-proofed and coated; this is the most energy- and water-intensive stage.
- **Powertrain:** engines and gearboxes are made or assembled (or, for EVs, battery packs and motors), usually in a separate line or plant.
- **Final assembly:** the body and powertrain are 'married'; seats, wiring, glass and wheels are fitted on a moving line.
- **Quality and PDI:** tests, inspection and a pre-delivery inspection before the car goes to a dealer.

Major Indian plants include Maruti Suzuki's at Manesar and Gurugram, Tata's at Sanand, Mahindra's at Chakan and Hyundai's near Chennai.* Plant capacity, shifts and how fast a model can be sold set a car's waiting period.

3. The supply chain

A car has around 30,000 parts.* The maker designs, assembles and brands the car, but buys most parts from suppliers. **Tier 1** suppliers (seats, electronics, transmissions) deliver complete systems to the plant; **Tier 2** and **Tier 3** supply components and raw materials to them. A shortage anywhere stops the line.

- **Localisation:** the share of parts made in India. Higher localisation cuts import duty, currency risk and cost, and is a stated goal of programmes such as the government's production-linked incentive (PLI) scheme for the auto sector.*
- **Semiconductors:** the chip shortage of 2021–22 forced several makers to cut production and extended waiting periods.*
- **EV-specific risks:** battery cells and rare-earth magnets (used in many EV motors) are largely imported, so EV costs and supply are exposed to overseas policy and prices.*
- **Just-in-time:** many suppliers deliver hours before parts are needed, which keeps stock low but makes disruptions costly.

4. How to read a spec sheet

A spec sheet is a list of numbers; the skill is to ask what each one tells you. Use the Swift's figures from earlier parts as an example.

Number	What it tells you	Swift example and what to watch
Engine cc, cylinders, aspiration	Size and style of the engine (Part 1)	1,197 cc, 3 cylinders, natural. Compare power per litre, not cc.
Power (bhp @ rpm)	Top-end performance (Part 2)	80.46 bhp @ 5,700 rpm: you must rev to use it.
Torque (Nm @ rpm)	Pulling strength and its spread	111.7 Nm @ 4,300 rpm: a narrow, higher-rpm peak.
Kerb weight	Power-to-weight, efficiency	About 925 kg (listing)*, so about 87 bhp per tonne.
Transmission	How it drives (Part 3)	5-speed manual or 5-speed AMT.
ARAI mileage	Comparison tool, not a promise (Part 2)	Expect real figures well below the claim.
Fuel tank	Range = tank × real km/l	37 litres (listing).
Boot space	Luggage practicality	265 litres (listing).
Ground clearance, turning radius	Rough-road ability, city manoeuvres (Part 4)	Check both on the brochure.

Number	What it tells you	Swift example and what to watch
Airbags, ESC, NCAP score	Safety (Part 5)	Variant matters; check the tested version.

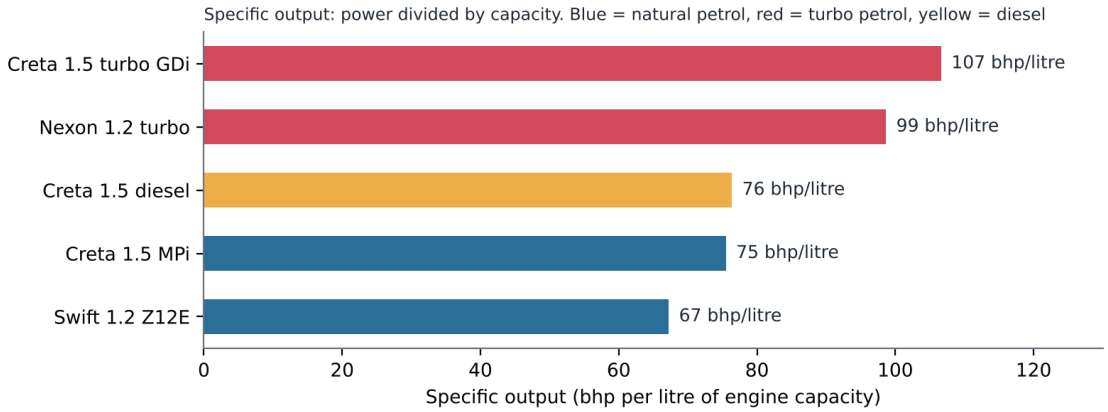


Fig 2. Specific output, power per litre of capacity, for engines in Part 1. Boosted engines extract 40–50% more power per litre than natural petrols.

Two derived numbers are worth computing for any car. **Specific output** (power ÷ litres) compares engines of different sizes: the Swift manages 67 bhp per litre, the Nexon 1.2 turbo 99 and the Creta turbo 107. **Power-to-weight** (power ÷ tonnes) compares how lively cars feel: the Swift's 87 bhp per tonne is a good benchmark for a small, light car.* Together with torque spread, these two numbers explain most of what you will feel on a test drive.

Summary

- A platform is the shared underbody; reusing it spreads fixed development costs across many cars.
- Break-even volume = fixed costs ÷ (price − variable cost); volume decides profit.
- A plant runs press shop, body shop, paint shop, powertrain, final assembly and quality checks.
- Supply chains run in tiers; chips, rare-earth magnets and battery cells are the big risks.
- Read a spec sheet by asking what each number implies; compute specific output and power-to-weight.
- Always compare the exact variant: airbags, gearbox and weight differ by trim.

*Unverified or general knowledge: the TNGA GA-C platform name for the Hycross, ALFA-ARC and acti.ev details, the plant locations, the number of parts in a car, the PLI scheme description, the 2021–22 semiconductor shortage and EV-component import exposure, and the Swift's kerb weight (from a listing) were not checked against official documents. The ₹2,000 crore platform cost and volumes are made-up numbers to show the arithmetic. Cross-checked in November 2026: INGLO (BE 6, rear motor), MQB-A0-IN (Kushaq, Slavia, Taigun, Virtus) and Heartect (Dzire). Spec-sheet figures: Wikipedia and Autocar India listings.

See also: the rest of the Automobile Engineering series (parts 1 to 12). Next: Part 10.

AUTOMOBILE ENGINEERING · PART 10 OF 12

The Indian Car Market Today

By Isagi · 9 November 2026 · All specs as of November 2026 (manufacturer brochures and listings)

TL;DR

India sold about 47 lakh passenger vehicles in FY2026 (April 2025 to March 2026). Official SIAM wholesale was 46,43,439 units (+7.9%); FADA retail registrations were 47,05,056 (+13.0%). Maruti Suzuki still sells nearly two in five cars (about 39% share), but its share is slipping as Mahindra, Tata, Toyota and Skoda grow and Hyundai shrinks. The market is concentrated: three makers hold two-thirds of sales. Utility vehicles (SUVs plus MPVs) were 67% of wholesale. Electric cars reached 4.25% of FY2026 retail and were 7–9% of monthly sales by September–October 2026. As of November 2026, the numbers below are verified against SIAM, FADA and trade compilations; sources differ slightly.

1. How big is the market, and which number to use

Two data sets are used in India, and they answer different questions:

- **Wholesale (dispatches):** cars the maker sent to dealers, reported monthly by SIAM and company statements. SIAM's official FY2026 figure is 46,43,439 units, up 7.9%. A trade compilation by Rushlane gives 46,83,208 (+8.4% on 43.2 lakh in FY2025); the maker shares in Section 2 use that compilation, because SIAM does not publish maker-wise numbers in the same form.
- **Retail (registrations):** cars actually registered to buyers, from the government's VAHAN database via FADA. FY2026 came to 47,05,056, up 13.00% on 41.64 lakh. FADA's retail leaders: Maruti 18,68,386 (39.71%), Mahindra 6,31,638, Tata 6,13,513, Hyundai 5,78,337.

The two differ because dealer stock rises and falls. If makers dispatch more than dealers sell, stock builds up; wholesale grows faster than retail. A manager should read both: retail shows real demand, wholesale shows what the factories were doing.

2. Who sells what: FY2026 market share

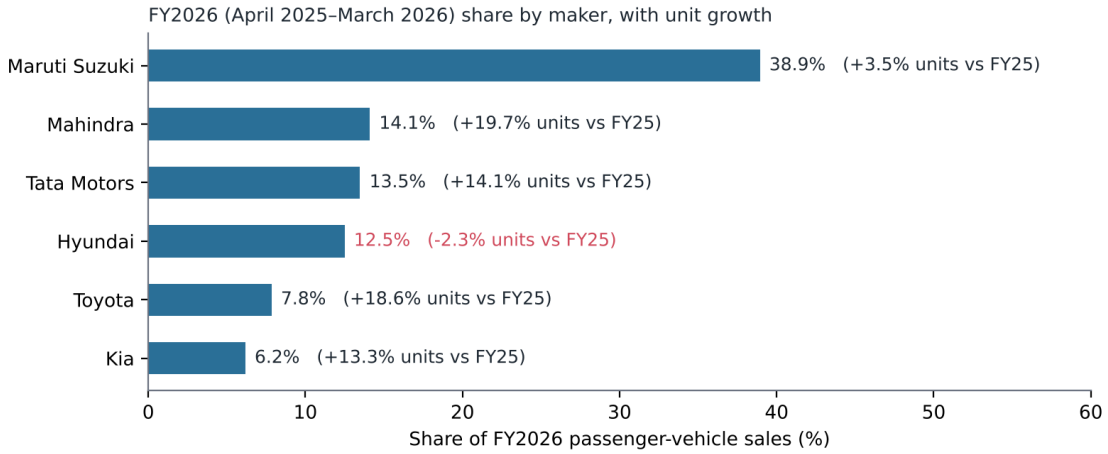


Fig 1. Share of FY2026 sales by maker (Rushlane wholesale compilation) with unit growth against FY2025.

Maker	FY26 units	FY26 share	Change in units
Maruti Suzuki	18,23,130	38.93%	+3.5%
Mahindra	6,60,276	14.10%	+19.7%
Tata Motors	6,31,387	13.48%	+14.1%

Maker	FY26 units	FY26 share	Change in units
Hyundai	5,84,906	12.49%	−2.3%
Toyota	3,67,152	7.84%	+18.6%
Kia	2,89,035	6.17%	+13.3%
Skoda	75,555	about 1.6%	+68.4%

Other FY2026 movements: MG grew 17.3%, Honda fell 6.0% and Volkswagen fell 11.0%. VinFast, a new entrant, had 2,390 retail registrations (FADA) covering part of its first year in India. Maruti's wholesale share fell from about 40.8% to 38.9%.

3. A concentrated market

- The top three makers (Maruti, Mahindra, Tata) hold 66.5% of the market; the top five add Hyundai and Toyota to reach 86.8%.
- A standard concentration measure, the Herfindahl-Hirschman Index (sum of squared shares), is about 2,150 from the six largest makers alone, a level that competition regulators in the US treat as highly concentrated (above 1,800).*
- In the retail data, Mahindra ranked second for the full year, about 18,000 cars ahead of Tata; in the January–June 2026 data, Tata (14.6%) was ahead of Mahindra (13.8%). Rankings flip between periods, so always state the period.

For a manager, the message is that scale advantages (platforms, dealer networks, supplier leverage) are large, and Mahindra and Tata, whose lineups are mostly SUVs, grew faster than the overall market (19.7% and 14.1% against 8.4%).

4. What actually sells: best-selling models

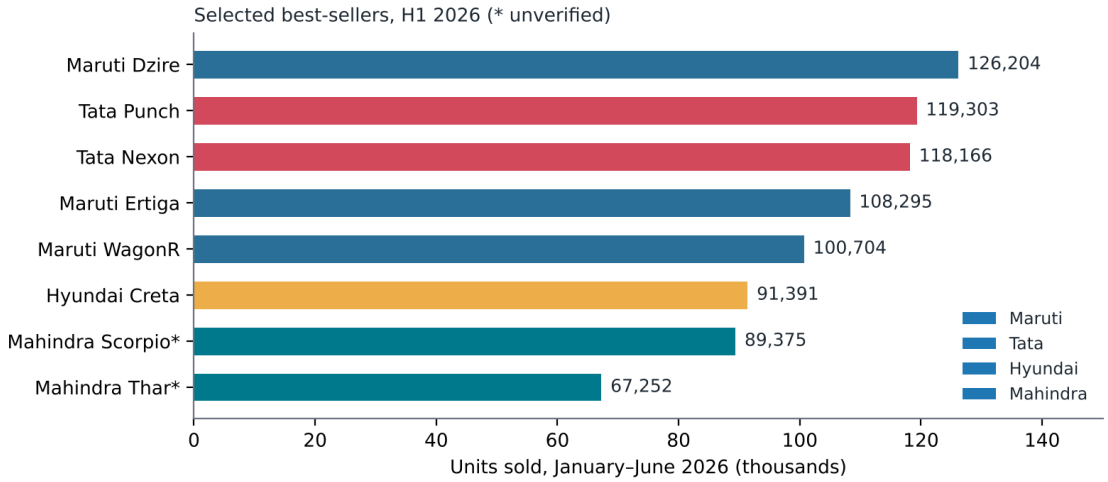


Fig 2. Selected best-sellers, January–June 2026 (H1 2026). The Scorpio entry combines the Scorpio-N and Scorpio Classic. * Scorpio and Thar figures could not be independently verified.

This is a selection, not a ranked top eight (Dzire, Punch, Nexon, Ertiga, WagonR and Creta are confirmed; Scorpio and Thar* are unverified, and other models may sit in between). Among these, five are SUVs (Punch, Nexon, Creta, Scorpio, Thar), one is a sedan (Dzire), one is an MPV (Ertiga) and one is a hatchback (WagonR). The Tata Punch topped annual sales in 2024 with 2,02,030 units, ending Maruti's long hold on the top spot. The picture moved again in April–September 2026 (H1 FY27): the Dzire (1,25,988) led the Punch (1,23,287) by only 2,701 units, followed by the Ertiga (1,12,557) and Nexon (1,11,919); the Creta sold 89,727 (−9.7%).

5. Forces shaping the market

- **SUV shift:** buyers keep moving up from hatchbacks to compact and mid-size SUVs. Utility vehicles (SUVs plus MPVs) were 67% of FY2026 wholesale, up from 65% in FY2025, and about 68% in the first quarter of FY2027.
- **Tax (GST 2.0):** from 22 September 2025, small petrol cars (up to 1,200 cc and 4 m) and small diesels (up to 1,500 cc) moved to 18% GST from 28%; larger cars moved to 40% with the compensation cess abolished; EVs stay at 5%; hybrids follow the same thresholds as petrol cars. This lowered the effective tax on small cars most. None of the sources I retrieved quantify its effect on sales.
- **Fuel choices:** EVs were 1,99,923 retail units in FY2026 (4.25%, up from 2.61%), CNG was 21.98%. In 2026 EV share was 8.5–8.7% in September and 7.3% in October (36,640 units), with hybrids at 9.94% in October; January–October EV registrations totalled 2,87,453. Parts 6 and 7 cover these.
- **Safety:** Bharat NCAP and Global NCAP ratings (Part 5) are now a marketing weapon, and six airbags are becoming common.
- **New entrants:** VinFast and Chinese-linked brands such as MG show that capital and product can disrupt a concentrated market.

6. How a manager should read these numbers

- Always note period (FY vs calendar half) and data type (wholesale vs retail).
- Share is zero-sum; look at units and growth to see who is actually growing.
- Look at models, not just brands: a few models carry a brand (Dzire, Punch, Nexon, Ertiga).
- Policy changes (GST, safety rules, fuel norms) move demand faster than product launches.

Summary

- India sold about 47 lakh passenger vehicles in FY2026 (SIAM wholesale 46.43 lakh, +7.9%; FADA retail 47.05 lakh, +13.0%).
- Maruti has about 39% but is losing share; Mahindra, Tata, Toyota and Skoda are growing, Hyundai is shrinking.
- The market is concentrated: the top three hold 66.5%.
- Compact SUVs dominate the best-seller list; utility vehicles are 67% of wholesale.
- GST cuts, safety ratings, hybrids/CNG and EVs are the main forces to watch.
- Always state the period and the data source when quoting market share.

*Unverified or general knowledge: the HHI threshold is the US antitrust guideline value and is applied here as a rough indicator; the Scorpio (89,375) and Thar (67,252) H1 2026 figures could not be independently verified. Cross-checked in November 2026: SIAM FY2026 wholesale (46,43,439, +7.9%), FADA FY2026 retail (47,05,056, +13.00%; Maruti 18,68,386, 39.71%), maker table (Rushlane wholesale compilation, 46,83,208), Dzire, Punch, Nexon, Ertiga, WagonR and Creta H1 2026 figures, April–September 2026 model ranking, Punch 2024 sales (2,02,030), utility-vehicle share, EV share and GST rates (PIB, India Briefing). Maker and period data differ across sources; units and shares may be revised.

See also: the rest of the Automobile Engineering series (parts 1 to 12). Next: Part 11.

How to Buy a Car

By Isagi · 9 November 2026 · All specs as of November 2026 (manufacturer brochures and listings)

TL;DR

Buying a car well is a sequence of decisions, not one: **budget** (on-road price and five-year cost, not the ex-showroom headline), **need** (body type and seats), **fuel and tech** (matched to your kilometres and charging access), **safety** (non-negotiables), a **shortlist of three** with real test drives, the right **variant**, then **finance, insurance and negotiation**. The numbers in this chapter use stated assumptions, so replace them with your own quotes. A key insight: over five years, depreciation (the fall in the car's value) is usually the largest single cost, bigger than fuel.

1. Start with the real budget

The ex-showroom price is not what you pay. Taxes and charges add roughly 15% or more.* The chart builds an on-road price for a ₹12 lakh ex-showroom car using assumed rates; yours will differ by state and variant.

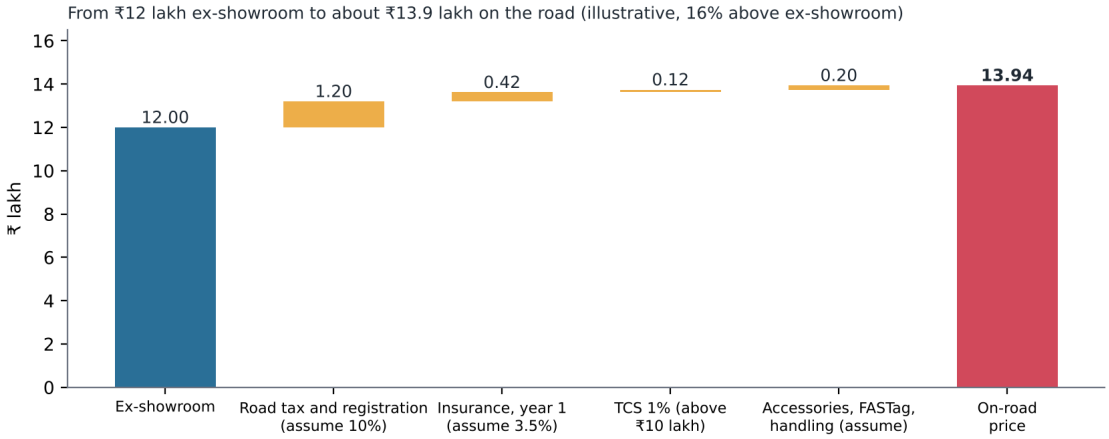


Fig 1. From ex-showroom to on-road price on assumed rates. Road tax varies widely by state and fuel type; insurance varies with the car's value and add-ons.

- **Road tax and registration:** paid to the state, varies a lot (some states waive it for EVs).*
- **Insurance:** third-party insurance is compulsory by law; comprehensive cover adds protection for your own car.
- **TCS:** tax collected at source of 1% applies on cars costing more than ₹10 lakh (Section 394 of the Income-tax Act, 2025; earlier Section 206C(1F)).
- **Extras:** accessories, extended warranty, FASTag, handling charges. Ask for an itemised quote and drop anything you do not want.

Then set a ceiling for monthly cost. A common personal-finance rule of thumb is that your car EMI should be no more than about 15% of take-home pay, and total car costs (EMI, fuel, insurance, service) no more than 10% to 15%.* Treat these as starting points, not laws.

2. Choose body type and size by need

- **Hatchback:** cheapest, easiest to park and run; best for city use and small families.
- **Sedan:** smoother ride and a separate boot; falling out of fashion but strong in value (the Dzire is the top seller in Part 10).

- **Compact SUV:** higher seating, ground clearance (Part 4), good for mixed Indian roads; the best-selling category.
- **MPV:** three rows, for large families or fleet use (Ertiga, Innova Hycross).
- **Larger SUV or 4x4:** highway comfort, space, off-road ability (Part 3); higher cost and running cost.

3. Choose fuel and technology by your kilometres

The right powertrain depends on how far you drive and what you can access. Use this table as a start, then do the break-even sum below.

If you...	Consider	Why
Drive under about 8,000 km a year	Petrol	Lowest purchase price; savings of other fuels rarely pay back
Drive 12,000 km or more, mostly in the city	Strong hybrid, CNG	Lower fuel bill per km (Part 7)
Can charge at home and drive mostly locally	EV	Cheapest per km (Part 6)
Drive long highway distances, carry loads	Diesel or efficient petrol	Torque and efficiency at steady speed (Part 2)

Break-even = extra price ÷ saving per km. Suppose the CNG version costs ₹90,000 more (an assumed premium) and saves ₹2 per km: you need 45,000 km to recover it, about 3.75 years at 12,000 km a year. If you will sell the car sooner, the premium may not pay back.

4. Set your safety non-negotiables

- Six airbags, ESC and ISOFIX child-seat anchors as standard on the variant you buy (Part 5).
- Check the NCAP rating for that variant, the protocol and the year.
- Treat ADAS as a bonus, not a substitute (Part 8).

5. Shortlist three, then test drive properly

Check	How	What it tells you
Ride over a speed breaker	20–30 km/h on a rough road	Suspension comfort and clearance (Part 4)
Pulling away and overtaking	From low speed, then at 60–80 km/h	Torque spread and gearbox response (Parts 2–3)
Traffic crawl	In stop-go conditions if possible	Clutch effort or automatic smoothness
Cabin and boot	Sit in the back; pack your usual luggage	Space and comfort for everyone
Noise and vibration	At idle and at 80 km/h	Refinement; three-cylinder vs four (Part 1)
Visibility, controls, screen	Park it	Daily usability

6. Pick the variant

Variants of one model can differ by lakhs of rupees. Base variants often lack safety and convenience items; top variants add features that cost more than they are worth to many buyers. A common approach is to choose the mid-level variant that has the safety items you need, the gearbox you want and the screen and features you will actually use.* Check that the airbag count and ESC match what you read about the model.

7. Finance, insurance and negotiation

A car loan's monthly instalment is: **EMI = P × r × (1+r)^n ÷ ((1+r)^n − 1)**, where P is the loan, r the monthly rate and n the number of months. For ₹8 lakh at 9% a year (an assumed rate), the EMI is about ₹25,440 over 3 years, ₹16,607 over 5 years and ₹12,871 over 7 years. The longer the loan, the lower the EMI but the more interest you pay:

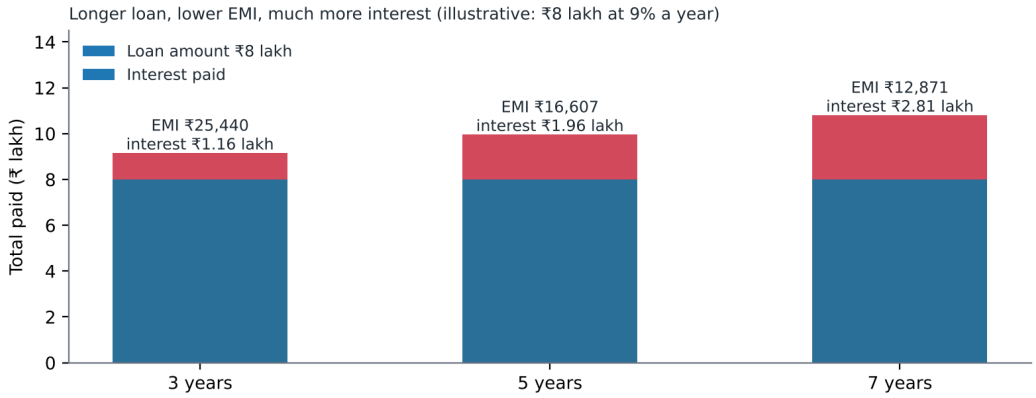


Fig 2. The same loan over three tenures. A 7-year loan pays about ₹2.8 lakh of interest, against about ₹1.2 lakh for 3 years.

- **Down payment:** a larger down payment cuts interest; avoid stretching tenure just to get a lower EMI.
- **Compare lenders:** bank, dealer and NBFC rates vary; ask for the full cost including processing fees.
- **Insurance:** compare comprehensive quotes from several insurers, not only the dealer's; consider a zero-depreciation add-on for a new car.*
- **Negotiate the whole deal:** ask for the discount, free accessories or exchange bonus in writing; check waiting periods and the delivery date.
- **Inspect at delivery (PDI):** check paint, panels, VIN, odometer reading, spare wheel and documents before you pay the balance.

8. The real cost of owning a car

Total cost of ownership (TCO) = on-road price + fuel + service + insurance renewals + interest − resale value. A five-year illustration, with every number assumed (60,000 km at ₹6 per km for fuel, 50% resale of the ex-showroom price):

Item (5 years)	₹ lakh	Note
On-road price	13.94	From Fig 1
Fuel	3.60	60,000 km × ₹6
Service and tyres	0.60	Assumed
Insurance renewals (years 2–5)	1.60	Assumed ₹0.40 lakh a year
Loan interest	1.96	₹8 lakh at 9%, 5 years
Less: resale value	−6.00	Assumed 50% of ex-showroom
Net cost of ownership	15.70	About ₹3.1 lakh a year, about ₹26 per km

In this example depreciation (on-road price minus resale value, ₹7.94 lakh) is about half of the net cost, while fuel is about one-sixth of gross outflows. That is why resale value matters as much as mileage when you compare two cars.* Cars with strong resale (often those from large, established brands) can cost less to own even if their price is higher.

Summary

- Budget on the on-road price and the five-year cost, not the ex-showroom headline.
- Choose body type by need, and fuel by annual kilometres using break-even = extra price ÷ saving per km.
- Make six airbags, ESC and ISOFIX non-negotiable on the variant you buy.
- Test drive three cars on real roads and check ride, pull, traffic comfort, space and noise.
- EMI falls as tenure rises, but interest paid rises sharply; keep the loan short.
- Depreciation is often the largest ownership cost; resale value belongs in the comparison.

*Unverified or general knowledge: the 15% on-road uplift, road-tax variation and EV waivers by state, the 15% and 10–15% personal-finance rules of thumb, the mid-variant heuristic, insurance add-on advice and the resale remark are general knowledge, not checked against official documents or your own quotes. The 1% TCS rule was cross-checked in November 2026 (TaxGuru, TDSMAN). Every rupee figure in this chapter (road tax 10%, insurance 3.5%, ₹0.20 lakh extras, 9% loan rate, ₹6 per km fuel, ₹90,000 CNG premium, 50% resale, service cost) is an assumption to show the method. The EMI formula is standard. Check current taxes and rates before deciding.

See also: the rest of the Automobile Engineering series (parts 1 to 12). Next: Part 12, the complete series in one PDF.

VERIFICATION LOG

What was checked, corrected and left open

A deep-research cross-check was run on 10 November 2026 against SIAM, FADA, Global NCAP and Bharat NCAP publications, manufacturer pages, the Press Information Bureau and trade press (Autocar India, Rushlane, Team-BHP and others).

Confirmed

- Engine and mileage figures for the Swift, Creta, Thar and Honda City (Parts 1 and 2).
- Nexon EV specifications and Skoda 1.0 and 1.5 TSI figures; Dzire Global NCAP result with a 4-star child rating.
- GST 2.0 rates from 22 September 2025, and the 5% rate for EVs.
- FY2026 market totals, H1 2026 model sales for the Dzire, Punch, Nexon, Ertiga, WagonR and Creta, and the 1% TCS on cars above ₹10 lakh.

Corrected

- Part 2: E20 is the standard pump fuel; official mileage loss estimates are 1–6%.
- Part 3: transmission entries for the Thar, Honda City and Skoda 1.5 TSI confirmed and the asterisks removed.
- Part 5: six airbags are not mandatory (the 2022 draft lapsed). Bharat NCAP dates and protocol updated; the crash-result table rebuilt with verified scores.
- Part 6: BE 6 relaunched as the Sporteq (59/70/79 kWh); no central purchase subsidy for private electric cars.
- Part 7: hybrid outputs restated in PS; Swift CNG power and torque corrected; E20 loss restated.
- Part 8: the ₹7.49 lakh XUV 3XO had no ADAS; the cheapest ADAS variant at launch was ₹11.99 lakh.
- Part 10: SIAM wholesale (46,43,439) separated from the trade compilation; chart relabelled 'selected best-sellers'; H1 FY27 update, SUV and EV shares added.
- Part 11: TCS now sits under Section 394 of the Income-tax Act, 2025.

Still unverified

- ABS and BS6 mandate dates; Level 3 autonomy and data-protection (DPDP) status for vehicles; road-tax and insurance ranges.
- Some specifications: BMW and Mercedes models, Innova Hycross, Nexon 1.2.
- Platform names and plant details (ALFA-ARC, acti.ev, TNGA GA-C, PLI), connected-car names, and the Scorpio and Thar H1 2026 sales figures.
- The Swift's Global NCAP star ratings and the Creta's Bharat NCAP status.

Prices, rates and sales figures change monthly. Check the manufacturer and official sources before you act on any number here.